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Scale Sensitivity and Regional Calibration of Full-Network Gauss–Newton Uncertainty in Physics-Informed Neural Networks

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Abstract

Physics-informed neural networks (PINNs) can be accurate on average yet inaccurate in thin layers and at moving interfaces. Useful uncertainty estimates should reflect these regional differences. We ask whether post-hoc Gaussian uncertainty from Gauss–Newton curvature, a linearised Laplace approach, places uncertainty where numerical errors occur. Its covariance depends on a prior scale and a common residual-noise scale. We evaluate full-network covariances for 17 Burgers and Allen–Cahn runs with two architectures, comparing reference-defined sharp and bulk regions. Allocation is strongly scale-dependent. For the trained prediction, five of eight Burgers–MLP runs have identified precisions matching sharp-to-bulk uncertainty and error ratios; these span more than four orders of magnitude. No match is found for three runs in the tested range. Neither training-based scale choices nor the tested predictive centres reliably calibrate both regions. Even at the matching points, a reference-informed oracle calibrating bulk coverage to 68% yields only 30.3–55.7% sharp coverage for the trained prediction. Reference-based readout refits reveal features capable of representing sharp structures. Spectral analysis explains how the scales redistribute uncertainty across curvature modes, with key bounds machine-checked in Lean 4. Matching regional uncertainty ratios therefore does not ensure calibrated intervals: allocation and coverage must be assessed separately.

Keywords: physics-informed neural networks, uncertainty quantification, Gauss–Newton covariance, calibration, spectral analysis

1 Introduction

Physics-informed neural networks (PINNs) approximate the solution of a partial differential equation (PDE) by minimising residuals of the governing equation, initial and boundary conditions, and available observations at sampled training points (Raissi et al., 2019). Competing loss terms and differential-operator conditioning can impede optimisation (Wang et al., 2021; Rathore et al., 2024); collocation strategies can also fail to propagate initial and boundary information into the interior (Daw et al., 2023). Converting a small sampled residual loss into a solution-error bound requires suitable stability and sampling conditions (Mishra and Molinaro, 2023). The remaining error can be highly non-uniform: a network may approximate the smooth bulk accurately while retaining substantial error near a thin layer or moving interface. Domain-averaged diagnostics can conceal this concentration.

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Uncertainty estimates could help identify these locally unreliable predictions, but their interpretation depends on what they describe. In scientific machine learning, uncertainty may arise from noisy observations, limited information about model parameters, or numerical approximation error in the computed solution (Psaros et al., 2023). These sources need not concentrate in the same parts of the domain. In particular, uncertainty under a residual-based training model need not track numerical approximation error. Bayesian PINNs combine physical constraints with a probabilistic model of noisy observations and propagate the resulting parameter uncertainty to the solution (Yang et al., 2021). Deep ensembles use variation between independently trained predictors (Lakshminarayanan et al., 2017). Whether this predictive spread tracks numerical approximation error must be assessed rather than assumed.

Local Gaussian methods construct uncertainty around a single trained network (Daxberger et al., 2021). A Laplace approximation uses the Hessian of the negative log-posterior at a posterior mode. For a Gaussian residual likelihood, the Gauss–Newton approximation replaces the negative log-likelihood Hessian by the noise-scaled residual Jacobian Gram matrix, omitting residual second-derivative terms. With a Gaussian prior, the resulting covariance is also the exact posterior covariance of the linearised (affine) residual model at the chosen checkpoint. Linearising the output propagates this parameter covariance to predictive variances (Immer et al., 2021b). In the related setting of neural operators, LUNO uses model linearisation to propagate Gaussian weight uncertainty into a function-valued Gaussian process (Magnani et al., 2025). This operator-learning setting differs from our fixed-solution residual model. Kristiadi et al. (2021) show that uncertainty geometry can change while point predictions remain fixed, motivating a separate assessment of covariance allocation.

The prior precision and residual-noise variance determine the relative strength of prior regularisation and residual constraints. Marginal likelihood, or evidence, integrates the likelihood over the prior and provides a training-based criterion for selecting these scales (Immer et al., 2021a). Antorán et al. (2022) account for the difference between a trained checkpoint and the affine posterior optimum using a fixed network linearisation. PINN applications already include eikonal travel-time inversion for hypocentre localisation with diagonal-Fisher curvature (Izzatullah et al., 2022), and the heat, wave and Burgers equations with full-Hessian Laplace inference and evidence-based selection of prior strength and residual weights (Graczyk and Witkowski, 2026). Our regional audit freezes the checkpoint and relative residual weights, leaving that joint training and hyperparameter-selection procedure outside the comparison.

Complementary approaches connect uncertainty more directly to numerical accuracy. Error-Aware B-PINNs incorporate numerical discrepancy into predictive uncertainty (Graf et al., 2022); solution-error bounds can set spatially varying likelihood widths in a second Bayesian fitting stage (Flores et al., 2025). Calibration assesses whether interval coverage agrees with its nominal level. Physics-informed conformal prediction uses PDE residuals as nonconformity scores to calibrate coverage in residual space without labelled solutions (Gopakumar et al., 2025). Conformal calibration of PINN predictions uses labelled solution values, with local quantiles allowing spatially adaptive bands (Yu et al., 2026). Marginal coverage guarantees, under their stated assumptions, do not by themselves ensure coverage conditional on membership in a sharp region. Section 4 illustrates this distinction with a split-conformal example. Our contribution is a controlled regional audit of a full-network residual Gaussian: scale and centre comparisons, reference feature refits and spectral analysis separate covariance allocation from interval coverage. Our audit asks how useful this covariance is for describing deterministic solution error under the specified likelihood.

We study two nonlinear parabolic problems: viscous Burgers, where nonlinear transport competes with diffusion and produces a thin layer, and Allen–Cahn, a reaction–diffusion equation with moving phase interfaces. The two architectures are an independent-feature multilayer perceptron (MLP) and its random-Fourier-feature variant. For each fixed trained network, we use the Gaussian posterior of the frozen affine residual and output model. Write $z = (x, t)$ for a space–time point, $u_\theta(z)$ for the trained prediction and $g(z) = \nabla_\theta u_\theta(z)$ for its parameter sensitivity. Let J be the Jacobian of the weighted training residuals and $F = J^\top J$.

For isotropic prior precision $\alpha > 0$ and common weighted-residual variance $v > 0$, the parameter covariance Σ and latent predictive variance σ^2 are

$$\Sigma = v(F + \gamma I)^{-1}, \qquad \sigma^2(z) = g(z)^\top \Sigma g(z), \qquad \gamma = \alpha v.$$

The prior variance is $1/\alpha$ in each parameter direction, and $\sigma(z)$ is the predictive standard deviation, without an added output-noise term. The common variance is applied after the fixed block weights. At fixed effective precision γ, the affine predictive centres and uncertainty contrasts are unchanged, while widths scale as $\sqrt{v}$. Our plug-in baseline uses $v = 1$ and $\alpha_0 = 1/\langle\theta^2\rangle$, with the mean square taken over all trainable weights and biases; this rule is not evidence maximisation. All these parameters enter the covariance. The Fourier frequencies are fixed embedding constants and are excluded from θ (Section 2).

Reference-gradient percentiles define the sharp and bulk regions, separated by a transition region (Definition 1). We compare sharp-to-bulk root-mean-square (RMS) uncertainty and error contrasts. Equal contrasts mean equal relative concentration, without determining absolute width or within-region coverage. The predictive centre is the prediction around which intervals are drawn. Because the prior is absent during training, the affine posterior mean can differ from the trained output. Comparing priors centred at zero and at θ isolates shrinkage towards zero from the likelihood correction. We also evaluate the shifted nonlinear network as a forward diagnostic with covariance frozen at θ. Reference solutions determine the regions and run admission, provide evaluation errors, and enter the explicitly reference-informed oracle and feature-refit diagnostics. At a specified checkpoint and scales, the covariance is formed from training residuals and network sensitivities.

All 17 admitted runs have a baseline allocation deficit: uncertainty is less concentrated in the sharp region than the error (Table 2; Figure 1). This relative deficit coexists with substantial regional overcoverage. The training-based scale choices change both properties: with trained centres, the conditional noise maximum-likelihood estimate under-covers both regions in all eight Burgers–MLP seeds, while finite-grid evidence selections frequently reach the lower noise boundary and can be numerically sensitive. Five Burgers–MLP seeds have identified contrast-matching precisions spanning more than four orders of magnitude; three retain the deficit throughout the sampled range.

Centring the prior at the checkpoint retains the baseline deficit in all 17 runs. Relative to the trained prediction, its affine mean improves full-grid accuracy in $15/17$ runs at the conditional noise-MLE scales, without calibrating both regions; nonlinear evaluation instead worsens bulk error in all 17. Even an oracle that matches trained-centre RMS contrasts and calibrates bulk coverage to 68% gives only 30.3–55.7% sharp coverage at the five identified Burgers–MLP matching points. Thus regional allocation and interval coverage are separate properties. Readout refits assess feature expressivity, while spectral analysis explains the scale dependence of covariance allocation.

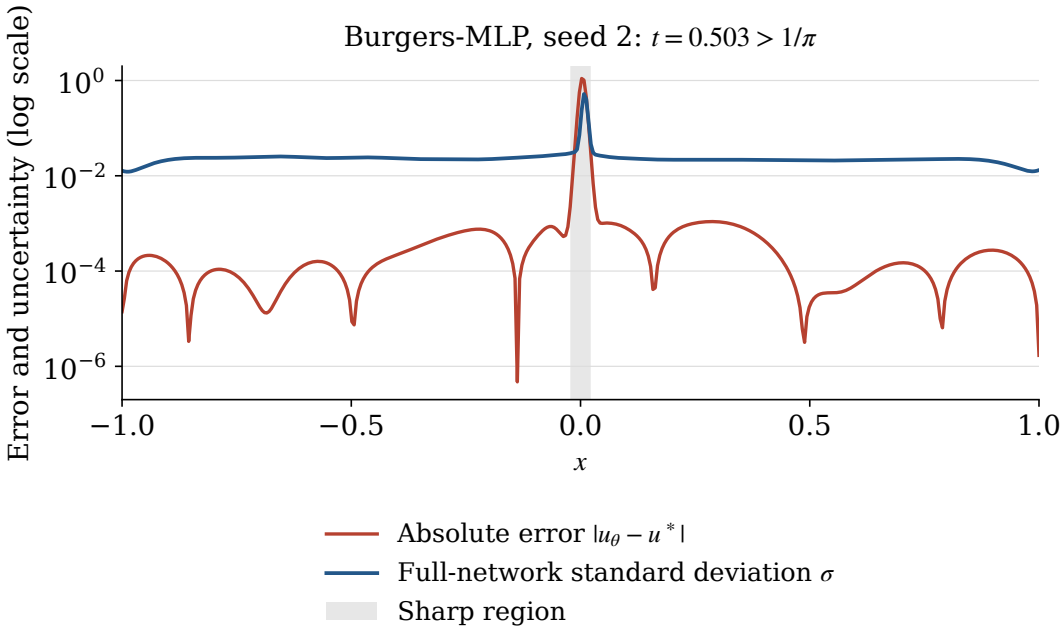


Figure 1: Burgers–MLP seed 2: absolute error and full-network baseline standard deviation at $t = 0.503$, the stored slice nearest $t = 0.5$, after breaking time $1/\pi$. The logarithmic scale is shared without rescaling; shading marks the global sharp region of Definition 1. Error is more concentrated near the layer. Here $|e| > \sigma$ near its centre; this is seed-specific, with post-formation calibration varying across seeds (Table 3). Full-grid contrasts are in Table 2.

Contributions.

- *Scale sensitivity and regional coverage (Section 3.1).* The tested scales and centres do not reliably calibrate both regions. The baseline deficit survives the alternative prior centre, and direct evaluation at refined contrast-matching precisions confirms that RMS agreement is insufficient after bulk calibration.
- *Reference feature expressivity (Section 3.2).* Using reference values to refit the final linear layer leaves below 4% of the trained sharp-region error energy on held-out time blocks in $16/17$ runs. This tests feature expressivity independently of residual-only coefficient selection.
- *Spectral weighting (Section 3.3).* A ratio of weighted spectral sums explains scale dependence. In the canonical runs, covariance weighting amplifies modest regional differences in total tangent norms. Two-subspace bounds constrain contrast but are too wide to certify the measured baseline deficit. Lean 4 verifies the spectral bounds and conditional scale limits, prior-centre bounds and coverage envelopes; reported run counts remain numerical.

Empirical scope. All eight attempted Burgers–MLP seeds are admitted. Burgers–Fourier, Allen–Cahn–MLP and Allen–Cahn–Fourier each have three admitted runs selected by the error

criterion in Section 2 and are exploratory. Attempted counts and rejected settings remain explicit. The results concern the specified training-weighted residual metric and reference-based partition.

2 Setting and Definitions

We specify the equations, admission rule and frozen affine model before comparing regional uncertainty under changes of scale and predictive centre.

Equations and trained networks. On $(x,t) \in [-1,1] \times [0,1]$, the viscous Burgers equation is

$$u_t + uu_x = \nu u_{xx}, \qquad \nu = 0.01/\pi, \qquad u(x,0) = -\sin(\pi x), \qquad u(\pm 1, t) = 0.$$

The positive-viscosity solution remains smooth and develops a thin layer near $x = 0$. We use the inviscid breaking time $1/\pi$ to define pre- and post-formation windows. The Allen–Cahn equation is

$$u_t = \varepsilon^2 u_{xx} + 5(u - u^3), \qquad \varepsilon^2 = 10^{-4}, \qquad u(x,0) = x^2 \cos(\pi x),$$

with periodic boundaries and interfaces of width $O(\varepsilon)$ between phases $u \approx \pm 1$. A contextual linear control is recorded in Appendix A; it is outside the nonlinear scale study.

The independent-feature MLP has 20 spatial and 20 temporal sub-networks feeding a fusion head, with $P = 196{,}009$ parameters. Replacing its input layer by random Fourier features gives $P = 275{,}369$ at bandwidth FS (Tancik et al., 2020). The parameter vector $\theta \in \mathbb{R}^P$ contains all trainable network weights and biases. The sampled Fourier frequencies are held fixed throughout training and uncertainty analysis and are excluded from θ. Both networks are trained in float64 by Adam (Kingma and Ba, 2015) followed by Levenberg–Marquardt (Levenberg, 1944; Marquardt, 1963). Evaluation uses a uniform 400×200 grid (80,000 points); the residual Jacobian has $N = 6400$ rows: 5000 collocation, 1000 initial-condition and 400 boundary rows. Training locations are sampled uniformly once per seed and held fixed; seeds jointly vary these locations, network initialisation and, for Fourier models, embedding frequencies. The Burgers reference uses the Cole–Hopf solution with 80-node Gauss–Hermite quadrature, agreeing with independent log-space quadrature to 4.5×10^{-14}. The Allen–Cahn reference uses fourth-order exponential time differencing with 512 modes and 1000 steps (Kassam and Trefethen, 2005); refinement to 2048 modes changes the canonical contrast by less than 0.1%. Appendix A records the training details. Unless a seed count is stated, numerical illustrations use Burgers–MLP seed 2 or Allen–Cahn seed 0.

Definition 1 (Region partition) *Let p_{70}, p_{95} be the 70th and 95th percentiles of $|u_x^*|$ over the whole evaluation grid. Define*

$$\mathcal{S} = \{|u_x^*| \geq p_{95}\}, \qquad \mathcal{B} = \{|u_x^*| < p_{70}\}, \qquad \mathcal{T} = (\mathcal{S} \cup \mathcal{B})^c.$$

These are the sharp, bulk and transitional regions.

The reference partition enters admission (Definition 2), evaluation and reference-informed diagnostics; it is absent from training and covariance construction. It follows moving and narrowing features: the Allen–Cahn interface moves by 58 grid cells, and sharp Burgers cells per slice fall from 58 at $t = 0.13$ to 8 at $t = 0.5$. No sampled reference-gradient magnitude equals either threshold. The top-5% definition alone does not establish resolution (Appendix A). For Burgers, the sharp

configuration	admitted/attempted	error range	median	τ
Fisher–KPP	$0/5$	81.7–170.5%	100.0%	2%
Nagumo, both initial conditions	$0/10$	75.9–90.3%	80.5%	2%
Kuramoto–Sivashinsky, $T = 100$	$0/5$	98.2–103.8%	100.4%	2%
Kuramoto–Sivashinsky, $T = 10$	$0/5$	9.9–17.0%	12.8%	2%
Burgers, MLP	$8/8$	0.05–0.40%	0.08%	2%
Burgers, Fourier	$3/8$	0.37–1.48%	0.78%	2%
Allen–Cahn, MLP	$3/5$	1.39–2.33%	2.33%	3%
Allen–Cahn, Fourier	$3/4$	1.64–2.60%	2.34%	3%
Convection–diffusion, MLP control	$3/4$	0.19–0.31%	0.21%	2%

Table 1: Admission accounting and relative L^2 error on $\mathcal{B} \cup \mathcal{T}$. Rejected rows include every attempted run; admitted rows include admitted runs only. Nagumo pools five seeds for each of two initial conditions. The control varies paired pulse width and diffusivity at one seed. Its narrowest MLP run is rejected at 62.2%; a separate Fourier attempt is rejected at 63.3%. All Kuramoto–Sivashinsky attempts also fail a relaxed 5% gate. Allen–Cahn MLP has median 2.330% and maximum 2.331%; both round to 2.33%.

threshold is 3.34, against maximum 116.8, and 69% of sharp points precede formation; temporal results therefore also use the two windows separately.

Definition 2 (Run admission) *A trained prediction is admitted if its relative error outside the sharp region satisfies*

$$\frac{\|(u_\theta - u^*)|_{\mathcal{B}\cup\mathcal{T}}\|_2}{\|u^*|_{\mathcal{B}\cup\mathcal{T}}\|_2} < \tau, \qquad \tau = 0.03 \textit{ for Allen–Cahn}, \quad 0.02 \textit{ otherwise}.$$

These trained-network decisions remain fixed in every scale and centre comparison. Excluding $\mathcal{S}$ avoids selecting directly on sharp error, but selecting a small bulk error can enlarge its regional ratio. Burgers–MLP has no seed filtering ($8/8$ admitted); the other nonlinear families are selected samples. A common 2% threshold retains $13/17$ runs; Appendix A reports the saved-output comparison on that subset.

Table 1 includes every attempted setting. A small training residual does not ensure an accurate field: one Kuramoto–Sivashinsky run has loss 4.4×10^{-5} at 103.8% relative error on $\mathcal{B} \cup \mathcal{T}$. The rejected settings therefore measure failure of this training-and-collocation protocol, consistent with known stiff-problem difficulties (Krishnapriyan et al., 2021; Wang et al., 2022).

Working likelihood, prior and centres. Write $r(\theta) \in \mathbb{R}^N$ for the stacked training residuals $r_{\mathrm{blk}} = w_{\mathrm{blk}} e_{\mathrm{blk}}$. The fixed factors are $w = 10$ on initial conditions, $w = 10$ on Burgers Dirichlet boundaries or Allen–Cahn periodic value matching, and $w = 1$ on collocation and periodic derivative matching. With common residual variance v, the working likelihood is $r(\vartheta) \sim \mathcal{N}(0, vI)$; hence the baseline negative log-likelihood is

$$-\log p = \tfrac{1}{2}\|r(\vartheta)\|^2 + \text{const} = \tfrac{1}{2}\sum_{\mathrm{blk}} w_{\mathrm{blk}}^2 \|e_{\mathrm{blk}}(\vartheta)\|^2 + \text{const} \qquad (v = 1).$$

The weights specify the likelihood as well as the training loss: the unweighted block noise standard deviations are $\sqrt{v}/w_{\mathrm{blk}}$. All analyses retain this raw weighted metric, with $J = \partial r/\partial\theta$ and $F = J^\top J$. For $\alpha, v > 0$, the isotropic prior is $\vartheta \sim \mathcal{N}(0, \alpha^{-1}I)$.

The baseline uses $v = 1$ and $\alpha_0 = 1/\langle\theta^2\rangle$, where $\langle\theta^2\rangle = P^{-1}\|\theta\|^2$ averages over all trainable weights and biases; this plug-in rule is not evidence maximisation.

Freeze the affine residual $r(\theta) + J(\vartheta - \theta)$ and affine output $u_\theta(x) + g(x)^\top(\vartheta - \theta)$, where $g = \nabla_\theta u_\theta$. In predictive expressions, x denotes the full space–time input z of Section 1; in the PDEs it denotes the spatial coordinate. Conditioning on zero residual gives

$$\Sigma(\alpha, v) = v(F + \gamma I)^{-1}, \qquad \sigma^2(x) = g(x)^\top \Sigma g(x), \qquad \gamma = \alpha v,$$

and the exact affine posterior mean

$$\delta_\gamma = -(F + \gamma I)^{-1}[J^\top r(\theta) + \gamma\theta], \qquad m_\gamma(x) = u_\theta(x) + g(x)^\top \delta_\gamma. \tag{1}$$

The prior is absent during training, so u_θ need not equal m_γ even at a stationary training solution. A θ-centred (checkpoint-centred) prior $\vartheta \sim \mathcal{N}(\theta, \alpha^{-1}I)$ instead gives

$$\delta_{L,\gamma} = -(F + \gamma I)^{-1} J^\top r(\theta), \qquad m^\theta_\gamma = u_\theta + g^\top \delta_{L,\gamma}.$$

Its covariance is unchanged. This prior is conditional on the fitted checkpoint. We compare its affine mean at the stated scales and available trained-centre roots, and evaluate $\widetilde{u}^\theta_\gamma = u_{\theta+\delta_{L,\gamma}}$ as a nonlinear forward diagnostic. The covariance remains at θ; this diagnostic is not a nonlinear posterior predictive mean. Prediction bounds are in Appendix C. At fixed γ, centres and uncertainty contrasts are unchanged by matched (α, v) choices, while widths scale as $\sqrt{v}$. No output-noise term is added when comparing latent uncertainty with the deterministic reference.

Baseline variances use the Woodbury identity

$$\sigma^2(x) = \alpha^{-1}\left[\|g\|^2 - (Jg)^\top (JJ^\top + \alpha I)^{-1} Jg\right].$$

The recorded baseline solves required no added diagonal regularisation. Weak-precision calculations instead factor J directly by QR/SVD to avoid cancellation (Appendix C).

Regional diagnostics. For each predictive centre, write e for prediction minus u^* and $\langle\cdot\rangle_r$ for a regional mean.

Definition 3 (Contrast, target and gap)

$$R^{\mathrm{rms}}_\sigma = \sqrt{\frac{\langle\sigma^2\rangle_{\mathcal{S}}}{\langle\sigma^2\rangle_{\mathcal{B}}}}, \qquad R_{\mathrm{tar}} = \sqrt{\frac{\langle e^2\rangle_{\mathcal{S}}}{\langle e^2\rangle_{\mathcal{B}}}}, \qquad \rho = R_{\mathrm{tar}}/R^{\mathrm{rms}}_\sigma.$$

The secondary mean-width contrast is $\widehat{R}_\sigma = \langle\sigma\rangle_{\mathcal{S}}/\langle\sigma\rangle_{\mathcal{B}}$.

Ratios assume positive denominators. At fixed scales, all centres share the contrast; the target, gap and coverage differ. The identity $\rho = 1$ matches regional RMS contrasts, not pointwise calibration or absolute width. We also report

$$\overline{Z}_r = \langle|e|/\sigma\rangle_r, \qquad \mathrm{cov@}k_r = \langle\mathbf{1}\{|e|/\sigma \le z_k\}\rangle_r,$$

configuration	admitted	R_σ^{rms}	R_{tar}	ρ
Burgers, MLP	$8/8$	6.37 ± 0.38	130.6 ± 82.4	20.5 ± 13.1
Burgers, Fourier	$3/8$	$[3.857, 3.959]$	$[15.77, 116.5]$	$[4.09, 29.55]$
Allen–Cahn, MLP	$3/5$	$[6.871, 7.731]$	$[21.58, 33.58]$	$[2.958, 4.888]$
Allen–Cahn, Fourier	$3/4$	$[5.818, 5.93]$	$[12.64, 17.37]$	$[2.165, 2.986]$
configuration	$\overline{Z}_{\mathcal{S}}$	$\overline{Z}_{\mathcal{B}}$	$\mathrm{cov@68}_{\mathcal{S}}$	$\mathrm{cov@68}_{\mathcal{B}}$
Burgers, MLP	0.319 ± 0.275	0.012 ± 0.003	0.923 ± 0.083	1.000 ± 0.000
Burgers, Fourier	$[0.3996, 0.724]$	$[0.02456, 0.1248]$	$[0.8227, 0.8912]$	$[0.9997, 1]$
Allen–Cahn, MLP	$[0.1845, 0.3296]$	$[0.02925, 0.03078]$	$[0.9708, 1]$	1
Allen–Cahn, Fourier	$[0.2533, 0.2893]$	$[0.02237, 0.0284]$	$[0.9958, 1]$	1

Table 2: Unit-noise, plug-in-precision baseline with trained centres. Burgers–MLP entries are mean $\pm$ population standard deviation over eight admitted seeds; other entries are observed ranges over three admitted runs. Counts are admitted/attempted. Ranges are descriptive, not confidence intervals. Calibration benchmarks are $\overline{Z} \approx 0.8$ and one-standard-deviation coverage ≈ 0.68.

with $z_{68} = 1$, $z_{95} = 1.96$. Coverage is the fraction of grid points whose intervals contain the deterministic reference, not a repeated-data coverage estimate; grid points are not independent replicates. Under a correctly specified centred Gaussian error model the expectations are $\sqrt{2/\pi} \approx 0.8$ and approximately $0.68, 0.95$. Throughout, sampled curves describe evaluated settings; refined roots locate the observed trained-centre crossings within retained numerical brackets. Direct root fields are compared with the coverage envelopes. A *best tested evidence setting* maximises a finite search, and a *consistency check* compares numerical representations. All spectral conclusions use the returned affine model; the checks do not certify weak-mode forward accuracy.

3 Results and Analysis

The three contributions concern regional allocation, last-layer expressivity and spectral weighting. We first compare uncertainty with realised error, then examine the features and curvature that determine these quantities.

3.1 Regional Allocation, Scale Sensitivity, and Predictive Centres

The comparisons retain the training-weighted homoscedastic likelihood, original reference regions and trained-network admission decisions. Burgers–MLP supplies eight unfiltered seeds; the other nonlinear families are smaller selected samples.

Regional allocation and scale sensitivity. Every admitted nonlinear run has a baseline allocation deficit (Table 2).

For canonical Burgers–MLP seed 2, $(R_\sigma^{\mathrm{rms}}, R_{\mathrm{tar}}, \rho) = (6.47, 118.58, 18.3)$. Across Burgers–MLP seeds, R_σ^{rms} varies by 6% (coefficient of variation), over $[6.00, 7.25]$, whereas $R_{\mathrm{tar}} \in [10.0, 238.2]$. All eight attempted seeds pass admission, so this family's deficit is not an admission-selection effect. Small bulk-error denominators amplify target variation; the bulk is also strongly over-dispersed at this baseline.

window	R_σ^{rms}	ρ	$\overline{Z}_{\mathcal{S}}$	cov@68	cov@95
$t < 1/\pi$	2.65 ± 0.23	16.9 ± 14.5	0.10 ± 0.07	0.99 ± 0.01	1.00 ± 0.00
$t \geq 1/\pi$	10.75 ± 0.71	20.6 ± 13.1	0.81 ± 0.81	0.76 ± 0.27	0.87 ± 0.17

Table 3: Burgers–MLP baseline split at $t = 1/\pi$, with trained centres; mean $\pm$ population standard deviation over eight seeds. Coverage and $\overline{Z}$ concern the sharp region. Pre-formation points constitute 69% of the global sharp mask.

The global mask mixes pre- and post-formation behaviour (Table 3). The mean gap exceeds one in both windows, so early points are not its sole source. Post-formation sharp coverage varies over $[0.37, 1.00]$ and $\overline{Z}_{\mathcal{S}}$ over $[0.02, 2.43]$, while $R_\sigma^{\mathrm{rms}} \in [9.93, 12.37]$: similar contrasts can accompany very different calibration. Allen–Cahn has no sharp points before $t \approx 0.4$ under the same global-percentile rule.

The baseline contrast also changes by at most 1.1% between the two finest evaluation subsamples (Appendix A); Gaussian and Student-t distributions with matched covariance have identical RMS and mean-width contrasts.

The scale sweep covers all 17 admitted nonlinear checkpoints with residual weights and reference regions fixed, using trained and zero-mean-prior affine centres. Burgers also uses the global masks intersected with $t \geq 1/\pi$. The unit-noise sweep contains 19 prior multipliers from 10^{-7} to 10^2 and matched points for the conditional noise MLE and finite-grid evidence selections. A centre change preserves R_σ^{rms} at fixed γ, but changes the target and calibration diagnostics.

Five trained-centre Burgers–MLP seeds cross $\rho = 1$ in the observed brackets (Figure 2). Saved modal sums refine the crossings to a range from 2.05×10^{-5} (seed 5) to 0.736 (seed 4), more than four orders of magnitude apart in effective precision (Table 12). The original sampled brackets are listed in Appendix C. Seeds 1–3 remain above one, with minimum sampled gaps 1.84, 2.59 and 1.99 (Appendix C). Seeds vary in both error target and weak-precision geometry: baseline uncertainty contrast varies by 6%, while each seed's largest sampled contrast ranges from 45.83 to 517.34. The QR-complement RMS ratios span $[42.71, 509.9]$ (Table 7). This complement is orthogonal to the retained N-column QR factor for $J^\top$ (Appendix C); its ratios identify the whole-kernel limit only under the rank condition in Section 3.3.

Training-based scale choices. The conditional MLE $\widehat{v} = \|r(\theta)\|^2/N$ fixes the trained parameter. Fitted-residual reuse can underestimate variance; this plug-in residual scale is not an unbiased discrepancy estimate. No classical degrees-of-freedom correction is assigned here (Appendix E). With trained centres it reverses the allocation deficit in seeds 0, 4 and 6; the other five have $\rho \in [1.30, 2.59]$. Nevertheless, all eight seeds under-cover both regions: one-standard-deviation coverage is 0.198–0.416 in bulk and 0.018–0.5315 in sharp; at 1.96 standard deviations it is 0.412–0.721 and 0.051–0.698.

On both zero-mean-prior evidence grids, $14/17$ best tested pairs lie at the lowest allowed noise variance; extending the range does not resolve those boundary selections. The three Allen–Cahn Fourier candidates are interior, whereas singular-value perturbations substantially change Allen–Cahn MLP selections. These are finite-grid candidates rather than reliably identified scales. Appendix C records the projected-data mechanism, extension and numerical sensitivity.

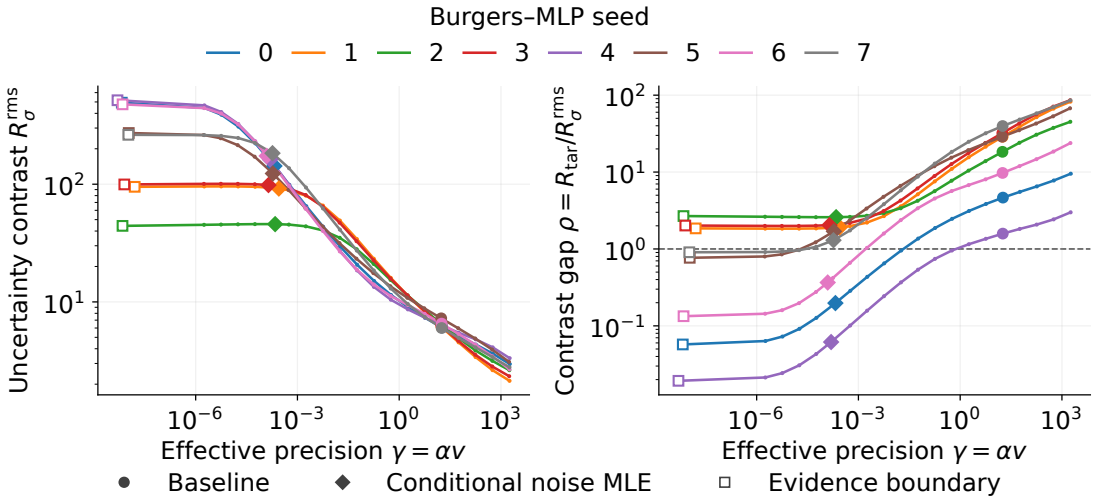


Figure 2: Uncertainty contrast (left) and gap $\rho = R_{\mathrm{tar}}/R_\sigma^{\mathrm{rms}}$ (right) for eight Burgers–MLP seeds with trained predictions fixed. Curves join 19 unit-noise prior settings and exact matched MLE/original-evidence points. Circles mark the baseline $(\alpha_0, 1)$, diamonds the conditional noise MLE $(\alpha_0, \widehat{v})$, and open squares the best original-grid evidence setting at the lower noise boundary. Selected settings share contrasts with their unit-noise aliases, but not absolute widths. The dashed line marks equal RMS contrasts, not calibration. Appendix C summarises other families, affine centres and extended-grid diagnostics; Appendix D gives their full curves.

Predictive centres and linearisation. At the baseline, the zero-mean-prior affine centre increases full-grid error RMS in $14/17$ checkpoints (Table 9). For Burgers–MLP, mean R_{tar} falls from 130.6 to 13.13 and mean ρ from 20.47 to 2.09, but mean sharp error RMS rises from 0.1138 to 0.2448 and bulk error RMS from 0.000805 to 0.01821; sharp error worsens in seven of eight seeds. Thus a reduced target can accompany worse prediction.

The displacement separates as $\delta_\gamma = \delta_{L,\gamma} + \delta_{P,\gamma}$, where the likelihood term is the shift under the θ-centred prior and $\delta_{P,\gamma} = -\gamma(F + \gamma I)^{-1}\theta$ is prior shrinkage. Its QR-complement term is $-\theta_c$ at every scale, where θ_c is the parameter projection onto that complement. At the baseline, total displacement is 99.78–99.96% of $\|\theta\|$, whereas the likelihood term is at most 0.0155%. Prediction-level bounds establish prior-only worsening in all 14 affected runs (Appendix C); parameter norms alone would not establish that attribution.

For the θ-centred prediction, the pointwise baseline shift bound ranges from 0.00749 to 0.0908 standard deviations. Bounds establish $\rho > 1$ in $16/17$; direct evaluation gives $\rho > 1$ in all 17, resolving Burgers–MLP seed 4 with gap 1.582. Both regions over-cover at one standard deviation. At the conditional-MLE scales, $\rho^\theta \in [2.18, 10.01]$. The affine mean improves full-grid RMS error in $15/17$ runs (Table 9), with median error ratio 0.7953. At one standard deviation, all 17 bulk regions over-cover and $12/17$ sharp regions under-cover. Thus prediction can improve without regional calibration. The residual variance is not re-estimated after shifting.

At the baseline, nonlinear evaluation $u_{\theta+\delta_{L,\gamma}}$ also retains $\rho > 1$ in all 17, with regional RMS affine remainder below 6.4×10^{-4} of RMS σ. At the MLE, however, the bulk remainder is 2.495–386.16 times RMS σ, despite $\|\delta_L\|/\|\theta\| \in [0.00782, 0.03867]$. Full-grid error worsens in $11/17$ and bulk error in all 17. Observed nonlinear gaps below one reflect bulk deterioration, not repaired allocation. These are forward diagnostics with covariance frozen at θ.

Oracle: contrast agreement and coverage. A descriptive oracle rescales each saved covariance to target 68% bulk coverage using the higher empirical quantile of $|e|/\sigma$, separately for each centre and time subset. It uses reference errors and is not a practical tuning rule. For the five trained-centre bracketing seeds, the closest saved points have $\rho \in [0.954, 1.032]$, yet sharp coverage is 0.343–0.652 at one standard deviation and 0.692–0.864 at 1.96. For zero-mean-prior affine post-formation predictions, the closest points across all eight seeds have $\rho \in [0.933, 1.193]$ and sharp one-standard-deviation coverage 0.009–0.347. At the five refined trained-centre roots, direct sharp coverage is 0.30325–0.55725 at one standard deviation and 0.67725–0.852 at 1.96 (Table 12). All values lie within the positive-modal envelopes, whose upper bounds are at most 0.563 and 0.862, respectively (Appendix C). Contrast agreement therefore does not restore sharp coverage under this bulk calibration.

At these five trained-centre roots, the θ-centred affine prediction has full-grid error RMS 0.493–0.973 times that of the trained prediction and $\rho^\theta \in [1.13, 7.58]$; these precisions do not match its own error contrast. After its own bulk-68% calibration, sharp one-standard-deviation coverage is 0.1165–0.424.

For $Z_j = e_j/\sigma_j$, define the variance-weighted regional second moment

$$M_r = \frac{\sum_{j\in r} \sigma_j^2 Z_j^2}{\sum_{j\in r} \sigma_j^2} = \frac{\langle e^2\rangle_r}{\langle \sigma^2\rangle_r}, \qquad \rho^2 = M_{\mathcal{S}}/M_{\mathcal{B}}.$$

Thus $\rho = 1$ equates weighted second moments, leaving ordinary coverage quantiles undetermined. At the five nearest saved oracle settings, using bulk-calibrated widths, sharp median $|Z|$ ranges over $[0.725, 1.273]$, compared with $[0.655, 0.701]$ in bulk; sharp 99th percentiles are 3.60–8.70, versus 3.05–4.66 in bulk. Sharp signed means include -1.96 (seed 5) and 0.85 (seed 7). The top 1% of sharp points carry 15.3–23.8% of error energy, and width–absolute-error Spearman rank correlations are 0.693–0.936. Larger sharp medians show that the difference extends beyond extreme tails; regional sign bias and substantial width–error ordering coexist with undercoverage.

An RMS deficit can coexist with overcoverage: seed 1 has closest gap 1.836 and sharp one-standard-deviation coverage 0.8605; seed 3 has gap 1.992 and coverage 0.725 (Tables 11 and 13).

Sharp calibration using its higher empirical 0.68 quantile gives sharp 1.96-standard-deviation coverage 0.8345–0.95775 for Burgers–MLP, below 0.95 in $7/8$ runs. Other families and time subsets include overcoverage (Table 13): matching one quantile need not match another. This output rescaling preserves within-region width patterns and does not test a residual-informed likelihood that changes $J^\top D^{-1} J$.

In the sampled band $0.9 \le \rho \le 1.1$, Burgers–MLP seed 7 has seven settings with sharp one-standard-deviation coverage 0.44725–0.986 after bulk calibration: $\rho = 0.925$ gives 0.756, while $\rho = 1.064$ gives 0.44725. These include undercoverage and overcoverage, without a continuous-band claim (Appendix C).

Ensemble comparison. An ensemble (Lakshminarayanan et al., 2017) of the eight Burgers–MLP runs, with spread computed at divisor $K = 8$, has $R_\sigma^{\mathrm{rms}} = 136.5$, its own mean-prediction target $R_{\mathrm{tar}} = 158.2$ and $\rho = 1.16$. Sharp $\overline{Z} = 0.64$ and coverage is $0.876/0.993$ at $68/95\%$,

while bulk $\overline{Z} = 0.45$. It nearly matches allocation but does not restore absolute calibration. The ordering against individual Gaussians depends on their scale: at the conditional MLE three trained-centre gaps are below one and five lie in $[1.30, 2.59]$, each relative to its own predictor's error. Omitting one ensemble member gives gaps 1.055–1.351, sharp 68% coverage 0.4935–0.9075 and 95% coverage 0.9720–0.9995. Omitting seed 5 gives the largest gap and lowest 68% coverage; its sharp error has correlation -0.7517 with the other seven's mean error. These ranges measure membership sensitivity, not confidence intervals: each omission changes the predictor, target and spread.

3.2 Feature Expressivity for the Reference Solution

The trained network's 129 last-layer features, including the bias, can approximate the sharp-region reference accurately when their readout is refitted using reference values. This tests feature expressivity, separately from the coefficients selected by residual-only training or the affine posterior. Let Φ be the feature matrix on the $80{,}000$ evaluation points and $e = u_\theta - u^*$ the sampled trained-network error.

Definition 4 (Per-region fraction and masked-global ratio) *For a region r with nonzero error e_r and local feature matrix Φ_r, let P_r denote the orthogonal projector onto $\mathrm{col}(\Phi_r)$ and define the* per-region *fraction*

$$f_r^{\perp} \;=\; \frac{\|e_r - P_r e_r\|^2}{\|e_r\|^2}. \tag{2}$$

Let P instead denote the projector onto $\mathrm{col}(\Phi)$ formed once globally, and define the masked-global *ratio*

$$\tilde{f}_r^{\perp} \;=\; \frac{\big\|(e - Pe)\big|_r\big\|^2}{\|e_r\|^2}.$$

The linear readout gives $u_\theta|_r \in \mathrm{col}(\Phi_r)$. Since $e_r = u_\theta|_r - u^*|_r$,

$$(I - P_r)e_r = -(I - P_r)u^*|_r, \qquad f_r^{\perp} = \frac{\|(I - P_r)u^*|_r\|^2}{\|e_r\|^2}.$$

The numerator is local reference-approximation error; the denominator is trained-network error. Their regional ordering therefore depends on both.

Remark 5 (The per-region diagnostic is a fraction) *For a self-adjoint idempotent P, orthogonality gives $\|e\|^2 = \|Pe\|^2 + \|e - Pe\|^2$, so $f_r^{\perp} \in [0, 1]$ for $e \neq 0$.*

Remark 6 (The masked-global quantity is not a fraction) *Global orthogonality need not survive restriction. A counterexample uses $\varphi = (1, 1)^\top$, $e = (1, 1/10)^\top$ and $r = \{2\}$. The projection $P_\varphi e = (11/20)\varphi$ gives $\tilde{f}_r^{\perp} = 81/4 = 20.25 > 1$.*

For comparison, $\tilde{f}_{\mathcal{B}}^{\perp} = 3.30$ on Burgers–MLP seed 2 and 4.4 on a synthetic control with error concentrated in a 5% subset.

Measured off-span fractions. Across all 17 runs, $f_{\mathcal{S}}^{\perp}$ is the smallest regional fraction. Its maximum is 0.011616 (Burgers–MLP seed 4): at least 98.8% of sampled sharp error lies in the local last-layer span. The bulk-to-sharp fraction ratio ranges from 9.37 to 542.5 (the displayed runs span

	Same grid: $f_r^{\perp}$			Held-out sharp region	
configuration	$\mathcal{B}$	$\mathcal{T}$	$\mathcal{S}$	random	10-slice blocks
Burgers, MLP (seed 2)	0.002	0.001	**0.0001**	0.000146	0.000455
Burgers, Fourier FS $= 2$ (seed 2)	0.071	0.057	**0.0015**	0.00163	0.00670
Allen–Cahn, MLP (seed 0)	0.230	0.030	**0.0009**	0.00139	0.0324
Allen–Cahn, Fourier FS $= 1$ (seed 0)	0.051	0.017	**0.0007**	0.000841	0.00987

Table 4: Reference expressivity of 129 last-layer features, including the bias. Residual energies are normalised by trained-network regional error. Same-grid columns give (2); held-out columns pool both test directions for random halves and alternating ten-slice time blocks.

18–266). Span containment makes this residual an upper bound for the full-network projection residual on the same points.

Held-out fits. Within each region we fit alternating ten-slice time blocks and test the remaining blocks, then swap, using unregularised least squares of numerical rank 129. The pooled test residual energy, divided by total regional error energy, is below 4% in $16/17$ sharp-region fits (median 0.72%). The worst fraction, 14.2%, is on Burgers–MLP seed 4, whose sharp error RMS 0.00362 is the smallest among Burgers–MLP seeds; its test residual RMS is 0.00136, versus error RMS 0.283 on seed 2. Random halves remain close to same-grid fits (Table 4), but blocked transition residual exceeds error energy in $10/17$ runs. Because the trained prediction u_θ lies in the last-layer span and each fit has full column rank, fitting e or u^* yields the same held-out residual energy. The pooled held-out residuals thus measure reference approximation by the local feature span, using additional reference information. These are local last-layer tests within the evaluation grid; span containment transfers only the same-grid bound, not held-out performance or a single global coefficient vector.

Expressivity and coefficient selection. Regional refits use reference values, whereas (1) selects coefficients from the residual likelihood and zero-mean parameter prior. The predictive-centre analysis in Section 3.1 attributes its baseline prediction worsening to prior shrinkage. Together, these diagnostics distinguish feature expressivity from coefficient selection; they do not identify the cause of the trained error or its uncertainty mismatch.

3.3 Spectral Weighting of Regional Variance

Covariance allocation is a weighted sum of tangent energies. Its modal quotient connects the scale curves to the trained features; coarse spectral bounds describe what remains identifiable after grouping those modes.

Modal weights and limits. Let $F = V\Lambda V^\top$, with eigenvalues $\lambda_i \geq 0$, largest eigenvalue $\lambda_{\max}$ and orthonormal eigenvectors v_i, and write $c_i(x) = g(x)^\top v_i$. At unit noise,

$$\sigma^2(g) = g^\top(\alpha I + F)^{-1} g = \sum_i \frac{c_i^2}{\lambda_i + \alpha}. \tag{3}$$

For residual variance v, each term is $vc_i^2/(\lambda_i + \gamma)$, $\gamma = \alpha v$. Defining $a_{r,i} = \langle c_i(x)^2 \rangle_r$ gives

$$(R_\sigma^{\mathrm{rms}}(\gamma))^2 = \frac{\sum_i a_{\mathcal{S},i}/(\lambda_i + \gamma)}{\sum_i a_{\mathcal{B},i}/(\lambda_i + \gamma)}. \tag{3a}$$

For these finite sums,

$$\lim_{\gamma\to\infty} R_\sigma^{\mathrm{rms}} = \sqrt{A_{\mathcal{S}}/A_{\mathcal{B}}}, \quad A_r = \langle \|g\|^2 \rangle_r, \qquad \lim_{\gamma\downarrow 0} R_\sigma^{\mathrm{rms}} = \sqrt{K_{\mathcal{S}}/K_{\mathcal{B}}}, \quad K_r = \langle \|\Pi_{\ker J} g\|^2 \rangle_r,$$

provided $A_{\mathcal{B}} > 0$ and $K_{\mathcal{B}} > 0$, respectively. Multiplication by γ gives both termwise limits. The QR-complement ratio equals the whole-kernel ratio when J has full row rank. For $A_{\mathcal{S}}, A_{\mathcal{B}} > 0$, at strong precision, writing $\mu_r = \sum_i a_{r,i}\lambda_i / A_r$ gives the more precise expansion

$$(R_\sigma^{\mathrm{rms}}(\gamma))^2 = \frac{A_{\mathcal{S}}}{A_{\mathcal{B}}} \left[1 - \frac{\mu_{\mathcal{S}} - \mu_{\mathcal{B}}}{\gamma} + O\left(\frac{\lambda_{\max}^2}{\gamma^2} \right) \right]. \tag{4}$$

This follows by expanding each $(\lambda_i + \gamma)^{-1}$ and dividing the two sums. Neither endpoint bounds intermediate contrast, and monotonicity need not hold. For example, $\lambda = (0, 1, 100)$, $a_{\mathcal{B}} = (1, 1, 100)$ and $a_{\mathcal{S}} = (1, 100, 1)$ give both limits equal to 1, but $R_\sigma^{\mathrm{rms}}(1) = \sqrt{10304/503} \approx 4.53$. The measured curves and strong-precision checks are in Appendix C. The conditional limits and counterexample are Lean-verified (Table 18); expansion (4) remains outside verification. Figure 2 changes the modal weights while holding the trained tangents fixed. The predictive centres share the covariance contrast but have different error targets.

Two-subspace bounds. We group the modes by a spectral cutoff.

Definition 7 (Spectral split and regional loadings) *At threshold $\epsilon > 0$, let Π_{sl} project onto modes with $\lambda_i \le \epsilon$ and Π_{st} onto those with $\lambda_i > \epsilon$. Define*

$$Q_{r,\mathrm{sl}} = \langle \|\Pi_{\mathrm{sl}} g\|^2 \rangle_r, \qquad Q_{r,\mathrm{st}} = \langle \|\Pi_{\mathrm{st}} g\|^2 \rangle_r.$$

For an individual vector, the lower and upper envelopes are

$$\begin{aligned} m_\epsilon(g) &= \frac{\|\Pi_{\mathrm{sl}} g\|^2}{\alpha + \epsilon} + \frac{\|\Pi_{\mathrm{st}} g\|^2}{\alpha + \lambda_{\max}}, \\ M_\epsilon(g) &= \frac{\|\Pi_{\mathrm{sl}} g\|^2}{\alpha} + \frac{\|\Pi_{\mathrm{st}} g\|^2}{\alpha + \epsilon}. \end{aligned} \tag{5}$$

Theorem 8, proved in Appendix B, bounds each pointwise variance between these envelopes. Averaging its termwise inequalities gives the primary regional bound

$$L_{\mathrm{reg}}(\epsilon) \le (R_\sigma^{\mathrm{rms}})^2 \le U_{\mathrm{reg}}(\epsilon). \tag{6}$$

Here $L_{\mathrm{reg}}(\epsilon) = \langle m_\epsilon(g) \rangle_{\mathcal{S}} / \langle M_\epsilon(g) \rangle_{\mathcal{B}}$ and $U_{\mathrm{reg}}(\epsilon) = \langle M_\epsilon(g) \rangle_{\mathcal{S}} / \langle m_\epsilon(g) \rangle_{\mathcal{B}}$, assuming positive total bulk energy. This is exact finite-sum aggregation, without a Jensen step. For common variance v, replace α by γ; the common factor cancels. The envelopes are sharp over spectra compatible with these summaries (Proposition 9, proved in Appendix B), but they discard energy placement within each block. They can therefore remain wide even after intersecting thresholds. Appendix B treats secondary aggregations; Appendix F, Table 18, records the formal verification scope.

Measured spectrum and loadings. With float64 machine epsilon $\varepsilon_{64} = 2^{-52}$, the canonical Gram eigensolves return positive eigenvalues spanning 19.6–21.2 decades, but values below $\varepsilon_{64}\lambda_{\max}$ are not individually resolved. The inverse lifts $J^\top w_i / \sqrt{\lambda_i}$, where w_i is a unit eigenvector of $JJ^\top$, are unstable near that scale. We instead resolve the separated strong subspace and compute

$$\|\Pi_{\mathrm{sl}} g\|^2 = \|g\|^2 - \|\Pi_{\mathrm{st}} g\|^2.$$

configuration	returned decades	below roundoff	$Q_{\mathcal{S},\mathrm{sl}}/Q_{\mathcal{B},\mathrm{sl}}$	$M_{\mathcal{S},\mathrm{sl}}/M_{\mathcal{B},\mathrm{sl}}$	RMS bracket
Burgers–MLP, seed 2	21.2	82%	259.5	514.9	$[0.0488, 307.86]$
Burgers–Fourier, seed 2	19.9	13%	43.7	195.4	$[0.0648, 103.51]$
Allen–Cahn–MLP, seed 0	20.4	66%	59.3	268.6	$[0.0982, 113.87]$
Allen–Cahn–Fourier, seed 0	19.6	26%	79.0	1132.4	$[0.1743, 55.92]$

Table 5: Canonical spectrum and baseline sub-α loadings. Returned decades include unresolved positive eigenvalues; the next column is the fraction below $\varepsilon_{64}\lambda_{\max}$. Primary loading uses $Q_{r,\mathrm{sl}} = \langle\|\Pi_{\mathrm{sl}} g\|^2\rangle_r$; the supporting mean-tangent loading uses $M_{r,\mathrm{sl}} = \|\Pi_{\mathrm{sl}}\langle g\rangle_r\|^2$. The final column gives (6) at $\epsilon = \alpha$, in contrast units, rounded outward from recorded energies. Only Burgers–MLP has replicated baseline loadings.

This includes Fisher-null directions without division by small eigenvalues. Orthonormality, cutoff separation, eigenpair error and independent variance checks are described in Appendix A.

Sharp-to-bulk tangent-energy ratios are close to one on Burgers: bulk/transition/sharp means are $134/136/141$ for MLP and $114/119/120$ for Fourier. The canonical strong-precision RMS limits are $1.02, 1.03, 1.46, 1.09$, whereas baseline configuration-mean contrasts are 3.92–7.30; covariance weighting supplies the amplification. At $\epsilon = \alpha$, the sharp field loads the weak subspace 43.7–259.5 times more strongly than bulk (Table 5). All 11 evaluated baseline brackets contain their measured RMS contrasts but are too wide to establish a deficit relative to the error target. For eight Burgers–MLP seeds, primary loading ratios span 171.2–268.0; supporting mean-tangent ratios span 380.6–742.0.

The full-grid study extends this geometry to all 17 runs. Among the full-time sampled curves, $13/17$ have both increasing and decreasing segments, and $12/17$ leave the interval formed by the tangent and QR-complement ratios. The saved fields reach $\gamma/\lambda_{\max} = 6.28 \times 10^{-5}$–$2.96 \times 10^{-4}$. Evaluation of their regional modal sums at $10^3\lambda_{\max}$ confirms approach to the tangent-norm limit, within 1.94×10^{-4} relative across all full and Burgers post-formation curves. Small positive singular values are retained without exact-null classification. The measured curves, conditional limits and coarse brackets answer distinct questions (Appendix C).

4 Discussion and Conclusion

Regional allocation and coverage are distinct properties of this post-hoc Gaussian. Likelihood and scales determine covariance allocation; the predictive centre also changes the error target and coverage. The tested training-only choices do not consistently calibrate both regions.

Predictive centre and linearisation. The baseline mismatch survives removal of zero-mean prior shrinkage. At weaker precision, affine prediction can improve without regional calibration, and a modest relative parameter shift can produce a large nonlinear remainder.

Held-out refits demonstrate feature expressivity, separately from coefficient selection. Spectral weights explain scale dependence, but the conditional limits do not bound intermediate contrast, and coarse envelopes cannot certify the baseline deficit. Evidence boundaries and perturbation sensitivity leave weak-precision scale selection unresolved.

Residual-informed likelihoods. Residual-informed or block-wise variances D replace F by $J^\top D^{-1} J$, after factoring out global noise variance, changing spectral directions and within-region allocation. The common-scale oracle does not test this model. The rejected Kuramoto–Sivashinsky example links small aggregate residuals to large solution errors, without ruling out useful local residual information. Such likelihood design remains untested.

Limitations. The study covers two $1{+}1$-dimensional nonlinear equations and two architectures with a shared sub-network/fusion-head design. Only Burgers–MLP supplies eight fully admitted seeds; the remaining families are small selected samples. Admission thresholds, raw residual weights and reference-based regions are experimental conventions. Root-coverage envelopes concern the five observed trained-centre crossings and the stated bulk calibration, rather than optimal joint calibration.

The θ-centred prior is conditional on fitted parameters. Nonlinear forward checks retain covariance at the original checkpoint; they do not establish a nonlinear posterior approximation. Spectral calculations do not certify weak-mode forward accuracy or exact rank. The synthetic evidence stress test in Appendix E.3 exceeds its whole-grid evidence tolerance, despite agreement at selected settings. Held-out expressivity is last-layer and within-grid; it does not establish off-grid or full-network held-out approximation. Transfer to higher dimensions, other architectures and training objectives is untested.

Implications. Assess accuracy, regional contrast and coverage for each predictive centre; ensemble sensitivity likewise warrants assessment of the particular ensemble. The trained-centre Gaussian is a covariance diagnostic. An archived random half-grid split-conformal illustration (Lei et al., 2018) on canonical Burgers–MLP seed 2, using absolute-error scores at target 0.95, gives marginal coverage 0.950 but bulk, transition and sharp coverage 0.98, 0.99 and 0.31. This descriptive finite-grid example shows how marginal coverage can conceal regional imbalance.

Code and data availability. Code, trained parameters, saved results and the Lean development are available at `https://github.com/soumya1729/pinn-uq-scale-sensitivity`. The tagged reproduction release supplies the saved-data archive and its checksum.

Conclusion. Under the specified training-weighted likelihood, regional covariance allocation is scale-sensitive and distinct from absolute interval calibration. The tested training-only scale rules and predictive-centre changes do not consistently calibrate sharp and bulk regions together, even when an affine mean improves accuracy. At the five identified trained-centre Burgers–MLP roots, sharp intervals under-cover despite bulk calibration and regional RMS contrast agreement. Reference-based readout refits show substantial sharp-feature expressivity, while spectral analysis explains how covariance weighting redistributes uncertainty. These findings support separate regional checks of prediction accuracy, uncertainty allocation and coverage for each run and predictive centre.

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Appendix A. Experimental and Reporting Details

The following details supplement the model and admission rule in Section 2. Training uses float64 throughout. Each architecture has 20 spatial and 20 temporal sub-networks of widths $[\,\cdot\,, 64, 64, 1]$ feeding a $[40, 128, 128, 1]$ fusion head. Activations are $\tanh$, biases are initialised to zero and weights are drawn from $\mathcal{N}(0, 2/d_{\text{in}})$. Fourier models use 16 random features per axis, with bandwidth 2 for Burgers and 1 for Allen–Cahn. Adam has learning rate 10^{-3}, with 5000 Burgers steps and $15{,}000$ Allen–Cahn steps, followed by 50 and 200 Levenberg–Marquardt steps. Each LM step uses at most 100 CG iterations, initial damping 10^{-4} and minimum 10^{-12}; it accepts at $\rho_{\text{LM}} > 0.25$, divides damping by 3 at $\rho_{\text{LM}} > 0.75$ and multiplies it by 4 on rejection, allowing at most 5 consecutive rejections. Residual row counts and weights are given in Section 2; complete implementation settings are preserved with the code.

Reported multi-run means average per-run quantities. Every $\pm$ denotes the population standard deviation with divisor n; ranges give observed minima and maxima. The three-run control varies paired pulse-width/diffusivity settings at seed 0. Mean gaps are means of ratios, not ratios of mean contrasts: on Burgers–MLP they are 20.47 and 20.49, respectively. Reference-error oracles are evaluation devices, not training-based tuning rules.

A common 2% admission cutoff retains $13/17$ of the original cohort: $8/8$ Burgers–MLP, $3/3$ Burgers–Fourier, Allen–Cahn MLP seed 3 and Allen–Cahn Fourier seed 1. The original 3% Allen–Cahn cutoff remains a convention, not a PDE-derived tolerance. The unchanged masks and saved fields permit the subset comparison below; its two Allen–Cahn results are single-run descriptions. In this subset, the baseline zero-mean affine error increases in $10/13$ runs; direct θ-centred baseline predictions give $\rho^{\theta} > 1$ in all 13, including the one case unresolved by the $12/13$ bound. With trained centres, conditional-MLE sharp coverage is below nominal in all 13 runs: $[0.018, 0.62425]$ at one sigma and $[0.051, 0.768]$ at 1.96 sigma. Bulk coverage is below nominal in $10/13$ and $12/13$ runs at those thresholds, respectively. All eight Burgers–MLP findings and the five refined trained-centre crossings remain in this subset by construction.

Linear-control context. The contextual linear control solves $u_t + au_x = \nu u_{xx}$ with $a = 1$ and a periodic Gaussian pulse. Three of four MLP attempts pass admission; the admitted seed-0 runs vary paired pulse-width/diffusivity settings. At the baseline their uncertainty contrasts are $[1.086, 1.331]$, error contrasts $[0.7816, 0.9986]$ and gaps $[0.6521, 0.899]$. They are not part of the nonlinear 17-run scale study.

Grid and spectral checks. The top-5% region rule alone does not establish feature resolution. For the sharpest slice, the discrete integral width in cells is

$$W_{\text{cells}} = \frac{\sum_j |\partial_x u^*(x_j, t)|}{\max_j |\partial_x u^*(x_j, t)|}.$$

This effective discrete width avoids the crossing-count floor of the half-maximum statistic; exceeding one cell does not establish continuum resolution. On nested spatial subsamples with recomputed masks, primary RMS and secondary mean-width contrasts change by at most 1.1% between $n_x = 400$ and 200. This sampling-density check concerns baseline contrasts, not convergence of oracle coverage. The canonical Gram spectral projectors were checked for orthonormality, separation at the cutoff and eigenpair backward error; independent Woodbury evaluations checked variances. The all-17 direct-J checks are summarised in Appendix C, with numerical protocols in Appendix E.

Appendix B. Spectral Bounds and Aggregation

We give the proofs supporting the two-subspace summaries in Section 3.3 and their aggregation to regional statistics.

Theorem 8 (Two-sided spectral sandwich) *For $F \succeq 0$, $\alpha, \epsilon > 0$ and $g_{\mathrm{b}} \neq 0$,*

$$L(\epsilon) := \frac{m_\epsilon(g_{\mathrm{s}})}{M_\epsilon(g_{\mathrm{b}})} \leq \frac{\sigma^2(g_{\mathrm{s}})}{\sigma^2(g_{\mathrm{b}})} \leq \frac{M_\epsilon(g_{\mathrm{s}})}{m_\epsilon(g_{\mathrm{b}})} =: U(\epsilon). \tag{7}$$

Consequently, the ratio lies between $\sup_{\epsilon>0} L(\epsilon)$ and $\inf_{\epsilon>0} U(\epsilon)$.

Proof In the weak block, $\alpha \leq \alpha + \lambda_i \leq \alpha + \epsilon$; in the strong block, $\alpha + \epsilon < \alpha + \lambda_i \leq \alpha + \lambda_{\max}$. Applying reciprocal bounds to (3) gives $m_\epsilon(g) \leq \sigma^2(g) \leq M_\epsilon(g)$. Both bulk envelopes are positive, so division in opposite directions gives the result. ■

The primary statistic averages variance before taking square roots. The secondary mean-width ratio is $\widehat{R}_\sigma = \langle\sigma(g)\rangle_{\mathcal{S}}/\langle\sigma(g)\rangle_{\mathcal{B}}$. Uniform pairwise bounds also transfer to it: if $\ell\sigma(g_b) \leq \sigma(g_s) \leq u\sigma(g_b)$ for every sharp–bulk pair, averaging over both finite regions and dividing by positive bulk mean width gives $\ell \leq \widehat{R}_\sigma \leq u$. This differs from evaluating variance at a region-mean tangent.

For regions with positive mean width, define $\kappa_r = \sigma(\langle g\rangle_r)/\langle\sigma(g)\rangle_r$. The covariance norm triangle inequality gives $0 \leq \kappa_r \leq 1$, and cancellation of the two mean widths gives, when $\kappa_{\mathcal{B}} > 0$,

$$\frac{\sigma(\langle g\rangle_{\mathcal{S}})}{\sigma(\langle g\rangle_{\mathcal{B}})} = \frac{\kappa_{\mathcal{S}}}{\kappa_{\mathcal{B}}}\widehat{R}_\sigma. \tag{8}$$

Equation (8) shows why mean-tangent loadings cannot replace the primary pointwise energies. The following sharpness result concerns the information in those energies.

Sharpness of the spectral bounds. The following construction shows what can be inferred from the four block energies alone. It does not assert that measured networks attain the bounding envelopes.

Proposition 9 (Sharpness) *Fix $\alpha > 0$, $0 < \epsilon < \lambda_{\max}$ and four nonnegative block energies, with positive total bulk energy. Spectra and vectors realising these energies can approach either endpoint of* (7)*.*

Proof For $\epsilon < \epsilon' \leq \lambda_{\max}$, use four orthogonal modes with eigenvalues $0, \epsilon, \epsilon', \lambda_{\max}$. To approach U, place the sharp weak and strong energies at $0, \epsilon'$, and the bulk energies at $\epsilon, \lambda_{\max}$. Square-root coordinates realise the prescribed energies and give

$$\frac{q_{\mathrm{s,sl}}/\alpha + q_{\mathrm{s,st}}/(\alpha + \epsilon')}{q_{\mathrm{b,sl}}/(\alpha + \epsilon) + q_{\mathrm{b,st}}/(\alpha + \lambda_{\max})} \longrightarrow U(\epsilon) \quad (\epsilon' \downarrow \epsilon).$$

For L, place sharp energies at $\epsilon, \lambda_{\max}$ and bulk energies at $0, \epsilon'$. The ratio then tends to $L(\epsilon)$. The strict inequality $\epsilon' > \epsilon$ keeps that mode in the strong block. ■

Appendix C. Full-Grid Scale and Centre Diagnostics

These diagnostics use all 17 admitted checkpoints: Burgers–MLP seeds 0–7, Burgers–Fourier seeds $0, 2, 6$, and Allen–Cahn MLP/Fourier seeds $0, 1, 3$. Trained parameters, raw weighted residuals, reference fields, 400×200 grids and global percentile masks remain fixed. Recentring does not change admission. Affine means use the frozen output; $u_{\theta+\delta_L}$ is a separate nonlinear diagnostic with covariance still at θ. The three-run families are exploratory. This appendix reports per-seed Burgers–MLP results; Appendix D gives extended curves and tables.

Burgers–MLP per-seed results. Table 6 lists the eight trained-centre runs. Its minima use the original grid and matched aliases, whereas the all-17 diagnostics also include expanded evidence and sensitivity-selected settings.

				bulk coverage		sharp coverage	
seed	baseline ρ	MLE ρ	min. sampled ρ	68%	95%	68%	95%
0	4.66	0.20	0.06	0.261	0.501	0.516	0.684
1	28.83	1.91	1.84	0.274	0.455	0.095	0.273
2	18.33	2.59	2.59	0.380	0.598	0.137	0.349
3	32.10	2.04	1.99	0.198	0.412	0.143	0.339
4	1.58	0.06	0.02	0.233	0.449	0.531	0.630
5	28.82	1.70	0.77	0.416	0.721	0.018	0.051
6	9.74	0.37	0.13	0.247	0.459	0.363	0.698
7	39.69	1.30	0.91	0.356	0.617	0.060	0.161

Table 6: Original Burgers–MLP scale study at the fixed trained centre. Baseline and MLE gaps use $(\alpha_0, 1)$ and $(\alpha_0, \widehat{v})$. Minima use the original 19 unit-noise prior multipliers and matched MLE/evidence points. Coverage is at the conditional MLE, within one and 1.96 standard deviations.

Approximate sampled crossing brackets in αv are

$$\begin{aligned} &\text{seed } 0: \ [0.0182150, 0.0576010], && \text{seed } 4: \ [0.569767, 1.80176], \\ &\text{seed } 5: \ [1.73345, 5.48164] \times 10^{-5}, && \text{seed } 6: \ [5.61912, 17.7692] \times 10^{-4}, \\ &\text{seed } 7: \ [1.78416, 5.64201] \times 10^{-5}. \end{aligned}$$

These intervals need not contain unique roots. No sampled crossing is found for seeds 1–3; this does not exclude unsampled crossings.

Settings and computation. The unit-noise sweep uses 19 prior multipliers 10^{-7}–10^{2}, with exact aliases for conditional-MLE and evidence-selected effective precisions. The evidence grids span $\alpha/\alpha_0 = 10^{-7}$–$10^{7}$ and initially $v/\widehat{v} = 10^{-4}$–10^{6} in quarter decades; one extension lowers the noise edge to $10^{-8}\widehat{v}$. Six prescribed spectral sensitivity cases supply further settings. For each prior centre, shared $\gamma = \alpha v$ gives the same contrast and affine mean; widths differ by $\sqrt{v}$, without added predictive noise.

One direct-J Householder QR/SVD per checkpoint gives $J^\top = Q_1 U \operatorname{diag}(s_i) V^\top$, with $Q = [Q_1, Q_c]$. All $N = 6400$ singular directions and $P - N$ complement coordinates are retained, without a Gram matrix, rank cutoff or noise floor. Prediction/loss reconstruction and baseline width

family	strong-γ ratio	QR-complement ratio	sampled R_σ^{rms}
Burgers–MLP	[0.9467, 1.023]	[42.71, 509.9]	[2.136, 517.3]
Burgers–Fourier	[1, 1.027]	[11.81, 30.54]	[1.361, 31.56]
Allen–Cahn–MLP	[1.457, 1.474]	[1.268, 2.532]	[1.465, 7.731]
Allen–Cahn–Fourier	[1.089, 1.138]	[25.21, 35.47]	[2.822, 31.08]

Table 7: Full-time regional RMS tangent-norm ratios, QR-complement ratios and measured contrast ranges across every saved γ and admitted run in each family. The QR column equals the whole-kernel weak limit when J has full row rank. Endpoint ratios are not global contrast bounds; the final column is a finite sampled range shared by all predictive centres.

comparisons give maximum width relative discrepancy 5.41×10^{-11}. At all 401 settings, independent differentiation checks $h_\gamma = J^\top(r + J\delta_\gamma) + \gamma(\theta + \delta_\gamma)$. The maximum normwise backward error is 6.19×10^{-17} against a 10^{-7} tolerance; complement norm, projection, output and null-membership checks have maximum scaled defect 5.10×10^{-16}. These consistency checks do not certify weak-mode forward accuracy or exact rank. Appendix E specifies their denominators.

The checkpoint-conditional θ-centred comparison uses the same fixed-scale covariance; Appendix E records its separate 95 checks.

Regional modal sums and conditional limits. Let $\lambda_i = s_i^2$, $q_{r,i} = \langle (U^\top Q_1^\top g(x))_i^2 \rangle_r$, $d_r = \langle \|Q_c^\top g(x)\|^2 \rangle_r$, and $A_r = d_r + \sum_i q_{r,i}$. Equation (3a) becomes

$$H_r(\gamma) = \frac{d_r}{\gamma} + \sum_i \frac{q_{r,i}}{\lambda_i + \gamma}, \qquad \langle \sigma^2 \rangle_r = v H_r(\gamma), \qquad R_\sigma^{\mathrm{rms}}(\gamma)^2 = \frac{H_{\mathcal{S}}(\gamma)}{H_{\mathcal{B}}(\gamma)}. \tag{9}$$

The conditional limits in Section 3.3 follow by multiplying (9) by γ: the strong limit requires $A_{\mathcal{B}} > 0$, and the finite weak ratio requires positive bulk energy in the whole kernel. The QR-complement ratio in Table 7 equals that weak limit if J has full row rank; tiny positive returned singular values do not certify this premise.

Neither limit bounds the intermediate quotient in general. Of the 17 full-time sampled curves, 13 have increasing and decreasing segments, and 12 leave the interval between the reported endpoint ratios. Allen–Cahn MLP has tangent/QR ratios about 1.27–2.53 but sampled maxima 7.30–7.73. The largest saved $\gamma/\lambda_{\max}$ is only 6.28×10^{-5}–2.96×10^{-4}; the saved modal sums also permit evaluation at stronger precisions without new fields. At $\gamma = 10^3 \lambda_{\max}$, all full and Burgers post-formation contrasts are within 1.94×10^{-4} relative to their tangent-norm limits. The first-order expansion (4) differs from the returned squared ratios by at most 3.43×10^{-7} relative to those squared ratios. Appendix D reports changed-scale brackets and loading ratios at $\epsilon = \gamma$; the brackets remain too broad to certify a realised-error deficit.

Evidence selection and mechanism. For the zero-mean prior's affine observation $y = J\theta - r$, set $z = V^\top y$ and $C_i = v + \lambda_i/\alpha$. The affine log evidence (MacKay, 1992) is

$$\ell(\alpha, v) = -\tfrac{1}{2} \sum_{i=1}^{N} [z_i^2/C_i + \log C_i + \log(2\pi)].$$

Fourteen original selections occur at $10^{-4}\widehat{v}$ and remain on the new $10^{-8}\widehat{v}$ boundary after extension; the three Allen–Cahn Fourier selections are interior and unchanged (Table 8). All selected alpha

family	runs	extended noise edge	selected $\gamma/\gamma_{\text{ext}}$
Burgers–MLP	8	8/8	$[0.562, 3.162]$
Burgers–Fourier	3	3/3	$[1, 1.778]$
Allen–Cahn–MLP	3	3/3	$[1.778, 3.16228 \times 10^5]$
Allen–Cahn–Fourier	3	0/3	1

Table 8: Zero-mean-prior evidence-grid outcomes and six-case sensitivity. The interval is perturbed selected effective precision divided by the extended selection, across each family's runs and cases. Perturbations change singular values and the affine observation coherently, keeping singular vectors fixed; fields at alternative selected scales use the original Jacobian. Appendix D gives full per-run values.

coordinates are interior. These zero-mean-prior selections are best tested grid pairs, not continuous optima.

For the θ-centred prior the same formula uses $z = -V^\top r$. On the extended grid, 14 selections lie at the lower noise boundary, two Allen–Cahn Fourier selections at the upper alpha boundary, and one is interior. Sharp 1.96-sigma coverage around its affine mean is below nominal in all 17 runs, ranging from 0.00275 to 0.90175.

Each case changes singular values by $\delta s = \varepsilon_{64} \max(N, P) s_{\max}$ or $10\delta s$, upwards, downwards or alternately, with zero clamping. It also changes z coherently by $(s' - s)\theta_{\text{modes}}$, where $\theta_{\text{modes}} = U^\top Q_1^\top \theta$. These are finite sensitivity cases, not certified error bounds. All Allen–Cahn MLP selections change in all six cases; their $1\times$ quadratic terms $\delta s^2/\gamma$ alone are 7.55, 488.55 and 228.65. Their extended selections are not treated as robust scale estimates.

At fixed alpha,

$$D_v = \frac{\partial \ell}{\partial \log v} = -\frac{1}{2} \sum_i \frac{v}{C_i} \left(1 - \frac{z_i^2}{C_i}\right). \tag{10}$$

At the fourteen extended boundary selections, the $\lambda_i/\gamma < 0.01$ group contains 603–4594 modes, with summed projected data energy only 0.000481–0.014164 times summed model variance (Figure 3 illustrates Burgers–MLP seed 2). Its negative slope contributions favour lower noise. Original/extended fixed-alpha slopes are -2633 to -2561 / -2379 to -2255 for Burgers–MLP, -481 to -417 / -407 to -317 for Burgers–Fourier, and -1503 to -1124 / -1397 to -753 for Allen–Cahn MLP. Allen–Cahn Fourier slopes are -291 to -113; an interior quarter-decade maximum need not have zero continuous slope. These fixed-alpha slopes describe local behaviour, not an alpha-reoptimised path or the limit $v \to 0$.

Synthetic validation. Appendix E reports dense posterior comparisons, correctly specified coverage draws, fitted-residual noise bias and evidence scale-recovery controls. The stressed 21.2-decade toy keeps the prescribed tolerance unchanged: selected pairs agree with its reference, but the whole-grid evidence discrepancy exceeds it. These controls do not establish PINN evidence stability.

Centre and regional displacement. Table 9 compares the prior centres and nonlinear diagnostic at fixed baseline/MLE scales; Table 10 reports relative regional displacements. A reduced error contrast need not improve accuracy. Covariance is shared, whereas errors and coverage are centre-specific. At the conditional MLE, all eight Burgers–MLP zero-mean-prior affine gaps lie

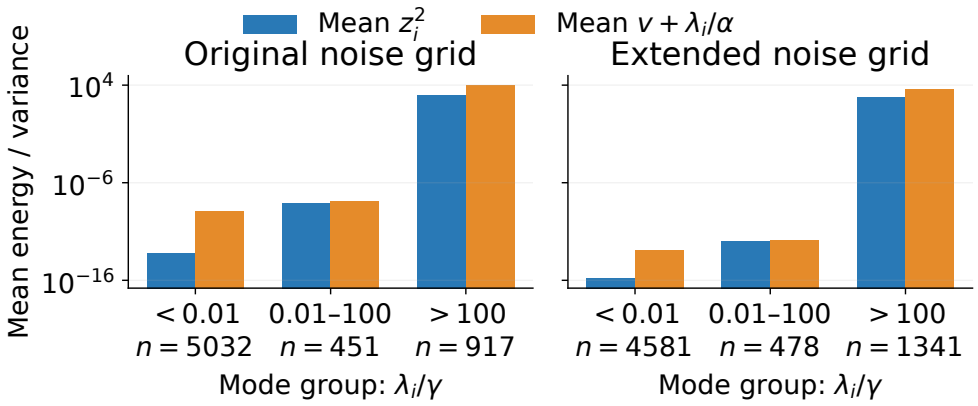


Figure 3: Burgers–MLP seed 2, zero-mean prior: projected affine-observation energy z_i^2 and model variance $v + \lambda_i/\alpha$ at the original and extended selections, averaged in three λ_i/γ groups. Counts accompany the bars. Little data energy in weak groups favours lower noise in (10); the grouping is descriptive.

			Baseline			Conditional MLE		
family	n	trained	zero	θ	nonlinear	zero	θ	nonlinear
Burgers–MLP	8	0.02546	$0.05883^{(7)}$	$0.02537^{(2)}$	$0.02537^{(2)}$	$0.023^{(5)}$	$0.02283^{(2)}$	$0.03944^{(6)}$
Burgers–Fourier	3	0.05124	$0.04316^{(1)}$	$0.05118^{(0)}$	$0.05118^{(0)}$	$0.04235^{(0)}$	$0.04935^{(0)}$	$0.05091^{(1)}$
Allen–Cahn–MLP	3	0.03287	$0.2776^{(3)}$	$0.03275^{(0)}$	$0.03275^{(0)}$	$0.02804^{(1)}$	$0.02455^{(0)}$	$0.2701^{(3)}$
Allen–Cahn–Fourier	3	0.03351	$0.245^{(3)}$	$0.03349^{(0)}$	$0.03349^{(0)}$	$0.03975^{(1)}$	$0.02496^{(0)}$	$0.03861^{(1)}$

Table 9: Paired full-grid RMS prediction errors. Entries are unweighted means of per-run RMS errors; superscripts count runs worse than the trained predictor out of n. Baseline uses $(\alpha_0, 1)$ and MLE uses $(\alpha_0, \widehat{v})$. Zero and θ denote the two affine prior-centre predictions; nonlinear is $u_{\theta+\delta_L}$, a forward diagnostic rather than a posterior mean. All comparisons retain the original covariance linearisation.

in $[1.51, 3.39]$; one-standard-deviation coverage is 0.9815–0.99998 in bulk and 0.4185–0.92475 in sharp. The trained-centre allocation reversals and bulk undercoverage therefore do not transfer to this centre.

Sampled oracle and post-formation coverage. At every saved γ, each centre/subset uses the higher empirical bulk-0.68 quantile of $|e_c|/\sigma$ as a width multiplier. The closest setting minimises $|\log \rho|$, with lowest γ breaking ties (Table 11). Trained Burgers–MLP seeds $0, 4, 5, 6, 7$ have full-time crossing brackets. Six zero-mean affine curves cross in each time subset; full-time seeds $1, 7$ and post-formation seeds $1, 3$ remain above one at every saved point.

Post-formation masks intersect the global masks with $t \geq 1/\pi$, giving $48{,}246$ bulk, 4928 transition and 1226 sharp points. Table 11 includes noncrossing runs; the closest sample need not have

family	region	baseline	MLE
Burgers–MLP	bulk	$[7.54, 54.1]$	$[0.264, 1.51]$
	sharp	$[0.58, 57.7]$	$[0.108, 8.55]$
Burgers–Fourier	bulk	$[3.87, 8.31]$	$[0.823, 2.42]$
	sharp	$[0.666, 0.864]$	$[0.167, 0.809]$
Allen–Cahn–MLP	bulk	$[15.7, 23]$	$[1.57, 2.59]$
	sharp	$[5.46, 7.22]$	$[0.416, 0.628]$
Allen–Cahn–Fourier	bulk	$[8.94, 9.95]$	$[0.97, 1.09]$
	sharp	$[4.38, 6.25]$	$[0.36, 2.26]$

Table 10: Across-run ranges of total regional RMS zero-mean-prior affine displacement divided by trained error RMS on the same full-time region. Baseline is $(\alpha_0, 1)$ and MLE is $(\alpha_0, \widehat{v})$. Displacement alone does not determine prediction accuracy; its alignment with trained error matters. Appendix D includes transition regions and evidence settings.

family	subset	trained 68%	affine 68%	trained 95%	affine 95%
Burgers–MLP	full	[0.3427, 0.8605]	[0.164, 0.9815]	[0.639, 0.9742]	[0.298, 0.995]
Burgers–MLP	post	[0.1639, 0.9959]	[0.008972, 0.3467]	[0.4176, 1]	[0.01387, 0.7121]
Burgers–Fourier	full	[0.1985, 0.5978]	[0.2828, 0.4662]	[0.4472, 0.7588]	[0.5145, 0.742]
Burgers–Fourier	post	[0.01713, 0.0938]	[0.08972, 0.3173]	[0.03589, 0.1819]	[0.1998, 0.5571]
Allen–Cahn–MLP	full	[0.0095, 0.0655]	[0.2015, 0.2755]	[0.0395, 0.2722]	[0.3588, 0.3962]
Allen–Cahn–Fourier	full	[0.1477, 0.2005]	[0.0775, 0.3827]	[0.2407, 0.4138]	[0.1432, 0.7338]

Table 11: Across-run sharp-region coverage ranges for trained and zero-mean-prior affine centres at each run's closest saved contrast setting, chosen separately for each centre and subset. A closest sample need not have $\rho \approx 1$, so these rows do not generally evaluate coverage after RMS-contrast matching. At each saved γ, a higher empirical bulk-0.68 quantile supplies the width multiplier; selection minimises $|\log \rho|$ and the lowest saved γ breaks exact ties. These reference-error oracles use all 8 Burgers–MLP or 3 other admitted runs, including runs without a crossing. "Post" uses the unchanged global masks intersected with Burgers $t \geq 1/\pi$. Bulk one-sigma coverage is about 0.68002; neither nominal sharp coverage nor an exact root is enforced.

$\rho \approx 1$. Table 12 instead gives direct evaluations at the five refined trained-centre roots, which do not match the other centres' error contrasts. Appendix D gives per-run curves.

Predictive bounds for the prior-centre comparison. For $\gamma, v > 0$, write $A_\gamma = F + \gamma I$. The zero-mean-prior shift splits as

$$\delta_\gamma = \delta_{L,\gamma} + \delta_{P,\gamma}, \qquad \delta_{L,\gamma} = -A_\gamma^{-1} J^\top r, \qquad \delta_{P,\gamma} = -\gamma A_\gamma^{-1} \theta.$$

In the factorisation above, their coordinates are $-s_i (V^\top r)_i / (s_i^2 + \gamma)$ and $-\gamma \theta_{\mathrm{modes,i}} / (s_i^2 + \gamma)$; the QR-complement contributions are 0 and $-Q_c Q_c^\top \theta$, respectively. The total norm includes their cross term. Replacing the prior by $\vartheta \sim \mathcal{N}(\theta, \alpha^{-1} I)$ removes $\delta_{P,\gamma}$ and leaves the covariance unchanged.

Define $\mathcal{K}_\gamma = r^\top J A_\gamma^{-1} J^\top r$ and $c_\gamma = \sqrt{\mathcal{K}_\gamma / v}$. Since $\delta_{L,\gamma}^\top A_\gamma \delta_{L,\gamma} = \mathcal{K}_\gamma$, Cauchy–Schwarz in the A_γ metric gives the pointwise bound

$$|m_\gamma^\theta(x) - u_\theta(x)| = |g(x)^\top \delta_{L,\gamma}| \le \sigma(x) c_\gamma, \qquad \mathcal{K}_\gamma = \sum_i \frac{s_i^2 (V^\top r)_i^2}{s_i^2 + \gamma}. \tag{11}$$

Let E_r and Σ_r denote trained error RMS and uncertainty RMS, and let $q_{r,i}$ be the regional mode energies in (9). The regional RMS triangle inequality also gives

$$B_r = \min\left\{ c_\gamma \Sigma_r, \sum_i |\delta_{L,\gamma,i}| \sqrt{q_{r,i}} \right\}, \qquad \max(0, E_r - B_r) \le E_r^\theta \le E_r + B_r. \tag{12}$$

For $E_{\mathcal{B}}^\theta > 0$ and $R_\sigma^{\mathrm{rms}} > 0$, $\rho^\theta \ge \max(0, E_{\mathcal{S}} - B_{\mathcal{S}}) / [(E_{\mathcal{B}} + B_{\mathcal{B}}) R_\sigma^{\mathrm{rms}}]$. The prior-only prediction differs from the archived zero-mean affine mean by $g^\top \delta_L$; its full-grid error is therefore at least $E_{\mathrm{full}}^{\mathrm{zero}} - B_{\mathrm{full}}$. This attributes worsening through a prediction-level bound rather than parameter norms. If $C_c(t)$ denotes empirical coverage at t standard deviations, then

$$C_{\mathrm{trained}}(t - c_\gamma) \le C_\theta(t) \le C_{\mathrm{trained}}(t + c_\gamma), \qquad t \ge c_\gamma,$$

by the pointwise triangle inequality.

The saved arrays reproduce all 401 displacement norms to 4.44×10^{-16} relative and regional variances to 6.73×10^{-14}. The full sweep has $\|\delta_\gamma\| / \|\theta\| \approx 0.9889$–$31.17$, which does not validate nonlinear linearisation. At the baseline, total displacement is $[0.997775, 0.999576]\|\theta\|$, its QR complement is $[0.976675, 0.985087]\|\theta\|$, and the likelihood-only norm ratio is $[8.73 \times 10^{-6}, 1.5413 \times 10^{-4}]$. Equations (11)–(12) give $c_\gamma \in [0.007489, 0.090775]$ and establish $\rho^\theta > 1$ in $16/17$ runs; seed 4 has inconclusive lower bound 0.4878. Direct evaluation resolves all 17 (Section 3.1). The same prediction-level bounds establish prior-only worsening in all 14 affected baseline cases. One-sigma coverage lower bounds are 0.7925 in sharp and 0.997196 in bulk, establishing overcoverage in all 17 runs. At the conditional MLE, $c_\gamma \in [75.428, 79.910]$ is uninformative; Table 9 reports direct evaluations.

Coverage envelopes at refined contrast-matching roots. At unit noise, each pointwise variance has the positive modal representation $q_\gamma(x) = \sum_i a_i(x)/(\lambda_i + \gamma)$, including the QR complement as a zero-eigenvalue mode, with $a_i(x), \lambda_i \ge 0$. For positive endpoint variances at saved precisions $0 < l \le \gamma \le h$, $l < h$, and $t = (\gamma - l)/(h - l)$, positivity gives

$$L_\gamma = \frac{(1-t) l q_l + t h q_h}{\gamma} \le q_\gamma \le U_\gamma = \left(\frac{1-t}{q_l} + \frac{t}{q_h} \right)^{-1}. \tag{13}$$

For the lower bound in (13), γq_γ is concave because every $\gamma/(\lambda_i + \gamma)$ is concave. For the upper bound, $q' = -\sum_i a_i/(\lambda_i + \gamma)^2$ and $q'' = 2\sum_i a_i/(\lambda_i + \gamma)^3$ satisfy $(q')^2 \le q q''/2$ by Cauchy–Schwarz. Hence $(1/q)'' = [2(q')^2 - q q'']/q^3 \le 0$, and taking the reciprocal of its chord gives U_γ. Both envelopes decrease with γ since $q_l \ge q_h$.

For a numerically refined root bracket $[a, b] \subset [l, h]$, use L_b and U_a to retain root-location uncertainty. The higher empirical bulk-0.68 quantile $Q_{\mathcal{B}}$ is monotone, so

$$k_{\min} = Q_{\mathcal{B}}(|e| / \sqrt{U_a}), \qquad k_{\max} = Q_{\mathcal{B}}(|e| / \sqrt{L_b})$$

		Sharp 1σ coverage		Sharp 1.96σ coverage	
seed	refined γ	direct	envelope	direct	envelope
0	0.02105515	0.3275	$[0.323, 0.333]$	0.78275	$[0.7815, 0.78475]$
4	0.73626974	0.427	$[0.41975, 0.43375]$	0.852	$[0.84525, 0.862]$
5	2.0476385×10^{-5}	0.30325	$[0.29775, 0.315]$	0.67725	$[0.6755, 0.67925]$
6	0.00163541	0.52325	$[0.5225, 0.526]$	0.847	$[0.8465, 0.84775]$
7	3.0590419×10^{-5}	0.55725	$[0.554, 0.563]$	0.8235	$[0.82125, 0.82675]$

Table 12: Direct sharp coverage and saved envelopes at five trained Burgers–MLP roots. Displayed precisions are rounded; root brackets have relative width below 5.3×10^{-13}. Bulk calibration uses the higher empirical 0.68 quantile. Envelopes include root and calibration uncertainty through (14); all direct values lie inside.

enclose the multiplier k_γ that calibrates bulk coverage at the root. For sharp coverage at threshold $z \geq 0$, this yields

$$\left\langle \mathbf{1}\{|e| \leq z k_{\min} \sqrt{L_b}\} \right\rangle_{\mathcal{S}} \leq C_{\mathcal{S},\gamma}(z) \leq \left\langle \mathbf{1}\{|e| \leq z k_{\max} \sqrt{U_a}\} \right\rangle_{\mathcal{S}}. \tag{14}$$

These analytic bounds use the returned spectral model and saved endpoint fields; they are not directed-rounding certificates for the original Jacobian. Direct pointwise root evaluations confirm the enclosures in Table 12. Root refinement concerns the five observed trained-centre brackets, without asserting uniqueness or global optimality. Appendix F maps the Lean verification of these analytic bounds. The evaluated run counts and refined brackets remain numerical results.

Regional calibration and sampled agreement band. At each trained-centre closest saved setting, use either $k_{\mathcal{B}} = Q_{\mathcal{B}}(|e|/\sigma; 0.68)$ or $k_{\mathcal{S}} = Q_{\mathcal{S}}(|e|/\sigma; 0.68)$, with the higher empirical quantile. Write $C^r_{\mathcal{S}}(z) = \langle \mathbf{1}\{|e| \leq z k_r \sigma\} \rangle_{\mathcal{S}}$ and $Z^r = e/(k_r \sigma)$ for the resulting coverage and standardised errors.

Table 13 gives full-grid per-seed results and family ranges, including the Burgers post-formation subsets. Sharp calibration fixes its one-standard-deviation coverage by construction: 0.68025 on 4000 full-grid sharp points and 0.68108 on 1226 post-formation points here. The informative outputs are coverage at 1.96 standard deviations and the absolute and signed standardised means. These are finite-grid descriptions, not repeated-draw guarantees. Coverage counts use inclusive comparisons $|e|/\sigma \leq z k_r$. Using the algebraically equivalent multiplied-width form can change a count by one grid point through floating-point rounding; all continuous summaries reproduce to relative tolerance 10^{-12}.

The band inventory contains all 28 run/subset settings with $0.9 \leq \rho \leq 1.1$ on full grids and Burgers post-formation masks. For Burgers–MLP seed 7, seven full-grid samples have $\rho \in [0.90635, 1.06370]$, with sharp bulk-calibrated coverage 0.44725–0.986 at one standard deviation and 0.78375–1 at 1.96. The other four crossing seeds each contribute one full-grid sample. This is a sampled inventory, not a continuous-band enclosure.

Width versus allocation. Changing γ reweights modes; changing v at fixed $\gamma = \alpha_0$ only rescales widths. Thus $\overline{Z} \propto v^{-1/2}$ and coverage is the empirical CDF of fixed standardised errors at $\sqrt{v}$. Figure 4 contrasts these operations; its fixed-γ curves are rescaled saved fields.

Burgers–MLP seed	ρ	$C_{\mathcal{S}}^{\mathcal{B}}(1)$	$C_{\mathcal{S}}^{\mathcal{B}}(1.96)$	$C_{\mathcal{S}}^{\mathcal{S}}(1.96)$	$\overline{\lvert Z\rvert}^{\mathcal{S}}$	$\overline{Z}^{\mathcal{S}}$
0	0.95438	0.34275	0.7885	0.92025	0.91339	−0.24502
1	1.8363	0.8605	0.97425	0.865	0.94283	0.85765
2	2.5874	0.44675	0.639	0.85025	1.3644	1.209
3	1.9924	0.725	0.933	0.9175	0.77179	0.56348
4	0.95486	0.437	0.86375	0.95775	0.83109	−0.069397
5	0.97331	0.35925	0.69175	0.8345	1.0436	−1.0371
6	1.0323	0.5165	0.8435	0.92825	0.85021	0.069544
7	0.96422	0.65175	0.837	0.84675	1.034	0.79421

Family / subset (n)	$C_{\mathcal{S}}^{\mathcal{S}}(1.96)$	$\overline{\lvert Z\rvert}^{\mathcal{S}}$	$\overline{Z}^{\mathcal{S}}$
Burgers–MLP / full (8)	[0.8345, 0.95775]	[0.77179, 1.3644]	[−1.0371, 1.209]
Burgers–Fourier / full (3)	[0.82475, 0.85075]	[1.0792, 1.3448]	[0.98518, 1.3234]
Allen–Cahn–MLP / full (3)	[0.921, 0.959]	[0.77708, 0.86039]	[−0.29925, 0.32371]
Allen–Cahn–Fourier / full (3)	[0.98325, 0.9995]	[0.71247, 0.81491]	[−0.36263, 0.27285]
Burgers–MLP / post (8)	[0.94454, 1]	[0.61868, 0.98575]	[−0.87154, 0.82267]
Burgers–Fourier / post (3)	[0.8646, 1]	[0.82852, 0.87527]	[0.7548, 0.86578]

Table 13: Trained-centre regional calibration at each run's closest saved contrast setting. The superscript $\mathcal{B}$ or $\mathcal{S}$ denotes the region whose higher empirical 0.68 quantile supplies the width multiplier. The upper panel gives eight full-grid Burgers–MLP seeds; the lower panel gives observed ranges for all 17 full-grid runs and the Burgers post-formation subsets. Each setting minimises $|\log \rho|$, with ties at the lowest saved precision. $\overline{\lvert Z\rvert}^{\mathcal{S}}$ and $\overline{Z}^{\mathcal{S}}$ are absolute and signed means of sharp-calibrated standardised errors. Sharp one-standard-deviation coverage is fixed by construction, up to discreteness and ties. Displayed values are rounded; reference calibration does not define a training rule.

Appendix D. Extended Empirical Records

The extended records use the fixed 17-checkpoint cohort, global reference masks and raw weighted residuals (Definitions 1 and 2). Covariance contrast is shared by the predictive centres at fixed scales; error contrast and the gap are centre-specific (Definition 3). "Full" denotes the complete evaluation grid. Burgers "post" intersects the unchanged masks with $t \geq 1/\pi$; no Allen–Cahn post-formation subset is imposed. Figures 5 and 6 show covariance contrasts and trained/zero-mean affine gaps; Table 9 adds the θ-centred comparison.

D.1 Evidence Selections and Regional Spectral Summaries

Table 14 extends the zero-mean-prior evidence summary in Appendix C. The initial evidence grid has lower noise edge $10^{-4}\widehat{v}$ and its one extension has lower edge $10^{-8}\widehat{v}$. The six perturbation cases are specified in Appendix E; selections are finite-grid candidates and displayed ranges are descriptive.

Changed-scale spectral partition. Let $q_{r,i} = \langle (U^\top Q_1^\top g)_i^2 \rangle_r$ and $d_r = \langle \|Q_c^\top g\|^2 \rangle_r$, using the factorisation in Appendix C. At $\epsilon = \gamma$, these saved energies give $Q_{r,\mathrm{sl}} = d_r + \sum_{\lambda_i \leq \gamma} q_{r,i}$ and $Q_{r,\mathrm{st}} = \sum_{\lambda_i > \gamma} q_{r,i}$. Substitution into Equation (5), with prior precision replaced by γ, gives the

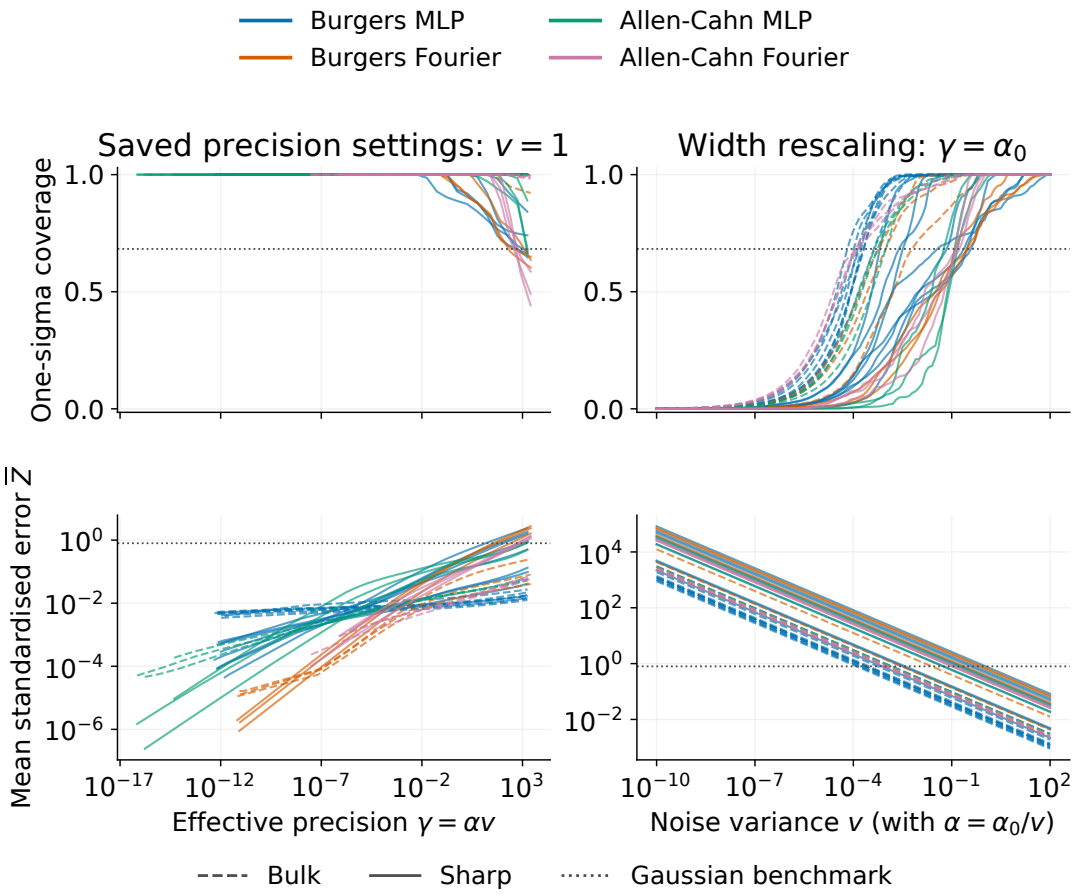


Figure 4: Trained-centre one-sigma coverage and mean standardised error for all 17 full-time runs. Colours identify families; dashed/solid curves denote bulk/sharp. Left: saved precisions at $v = 1$; right: width rescaling at $\gamma = \alpha_0$. Dotted lines mark Gaussian expectations 0.68269 and 0.79788. Lines join evaluated points without asserting confidence intervals.

returned-spectrum brackets in Table 15. Table 16 gives unweighted energy ratios. Neither certifies weak-mode forward accuracy.

D.2 Displacement under the Zero-Mean Prior

Table 17 extends Table 10 to transition regions and evidence settings. For the shift $c + b_\gamma$, define $c(x) = -g(x)^\top Q_c Q_c^\top \theta$ and $b_\gamma(x) = g(x)^\top Q_1 Q_1^\top \delta_\gamma$. The scale-independent c gives $\langle (c+b_\gamma)^2 \rangle_r = \langle c^2 \rangle_r + \langle b_\gamma^2 \rangle_r + 2\langle c b_\gamma \rangle_r$. Every full-grid sharp complement field contains both signs. For Burgers–MLP, its RMS is 0.001999–0.06048 and signed mean -0.02154–0.004895; $4/8$ baseline sharp cross terms are destructive. For Burgers–Fourier seed 0, complement sharp RMS 0.04236 exceeds total MLE shift RMS 0.04102. Component RMS values are nonadditive.

run	a_{orig}	a_{ext}	$v_{\mathrm{ext}}/\widehat{v}$	γ_{ext}	perturbed $\gamma/\gamma_{\mathrm{ext}}$	$\delta s^2/\gamma_{\mathrm{ext}}$
BM0	0.316	0.316	1×10^{-8}	6.64×10^{-13}	[1, 3.16]	0.0619
BM1	0.562	0.562	1×10^{-8}	1.59×10^{-12}	[1, 1.78]	0.0338
BM2	0.316	0.562	1×10^{-8}	1.24×10^{-12}	[0.562, 1.78]	0.0289
BM3	0.562	0.562	1×10^{-8}	7.87×10^{-13}	[1, 1.78]	0.0408
BM4	0.316	0.316	1×10^{-8}	4.95×10^{-13}	[1, 3.16]	0.0832
BM5	0.562	0.562	1×10^{-8}	1.06×10^{-12}	[1, 1.78]	0.0256
BM6	0.562	0.562	1×10^{-8}	6.99×10^{-13}	[1, 1.78]	0.0555
BM7	0.562	0.562	1×10^{-8}	1.03×10^{-12}	[1, 1.78]	0.0297
BF0	1	1	1×10^{-8}	6.71×10^{-12}	[1, 1]	0.00666
BF2	1	1	1×10^{-8}	8.52×10^{-12}	[1, 1.78]	0.00844
BF6	1	1	1×10^{-8}	9.23×10^{-12}	[1, 1]	0.00665
AM0	0.01	5.62×10^{-4}	1×10^{-8}	4.88×10^{-15}	[1.78, 1000]	7.55
AM1	0.00316	1.78×10^{-5}	1×10^{-8}	7.07×10^{-17}	$[17.8, 3.16\times10^{5}]$	489
AM3	0.00562	3.16×10^{-5}	1×10^{-8}	1.69×10^{-16}	$[5.62, 3.16\times10^{5}]$	229
AF0	0.562	0.562	0.01	8.03×10^{-7}	[1, 1]	4.83×10^{-8}
AF1	0.316	0.316	0.00178	3.11×10^{-8}	[1, 1]	1.63×10^{-6}
AF3	0.562	0.562	0.0562	4.85×10^{-6}	[1, 1]	6.6×10^{-9}

Table 14: Zero-mean-prior evidence selections and six-case sensitivity. BM/BF and AM/AF denote Burgers and Allen–Cahn MLP/Fourier, with seed suffixes; $a = \alpha/\alpha_0$. BM/BF/AM select the lower noise edges specified above; AF selections are interior and unchanged. The $1\times$ envelope δs gives the quadratic term, not a uniform eigenvalue error bound.

family	conditional MLE	extended evidence
Burgers–MLP	$[0.00491, 7.01\times10^{4}]$	$[0.00237, 1.88\times10^{5}]$
Burgers–Fourier	$[0.0258, 1.13\times10^{3}]$	$[0.0222, 1.43\times10^{3}]$
Allen–Cahn–MLP	[0.0079, 713]	$[0.00105, 3.06\times10^{3}]$
Allen–Cahn–Fourier	$[0.0161, 1.15\times10^{3}]$	$[0.0124, 1.97\times10^{3}]$

Table 15: Changed-scale two-block brackets at $\epsilon = \gamma$ from saved direct-J regional energies, in σ-ratio units. Intervals enclose the union of run-specific returned-spectrum brackets, rounded outward. Weak-mode forward accuracy remains uncertified, and the broad brackets cannot establish the realised-error deficit.

	$Q_{\mathcal{S},\mathrm{sl}}/Q_{\mathcal{B},\mathrm{sl}}$			$Q_{\mathcal{S},\mathrm{st}}/Q_{\mathcal{B},\mathrm{st}}$
family	MLE	original evidence	extended evidence	all three settings
Burgers–MLP	$[2.15\times10^{3}, 1.94\times10^{5}]$	$[1.95\times10^{3}, 2.67\times10^{5}]$	$[1.91\times10^{3}, 2.67\times10^{5}]$	[0.896, 1.05]
Burgers–Fourier	[150, 832]	[150, 998]	[145, 996]	[0.999, 1.05]
Allen–Cahn–MLP	[14, 20.5]	[4.59, 9.76]	[2.14, 7.02]	[2.12, 2.17]
Allen–Cahn–Fourier	[291, 693]	[476, 988]	[476, 988]	[1.19, 1.29]

Table 16: Changed-scale sharp-to-bulk tangent-energy ratios on the full grid, as across-run ranges. At $\epsilon = \gamma$, weak energy includes returned modes with $\lambda_i \le \gamma$ and the QR complement; stiff energy uses the remaining modes. The stiff ranges coincide across the three settings. These unweighted ratios are not uncertainty contrasts or weak-mode accuracy certificates. Evidence settings are best tested zero-mean-prior grid selections.

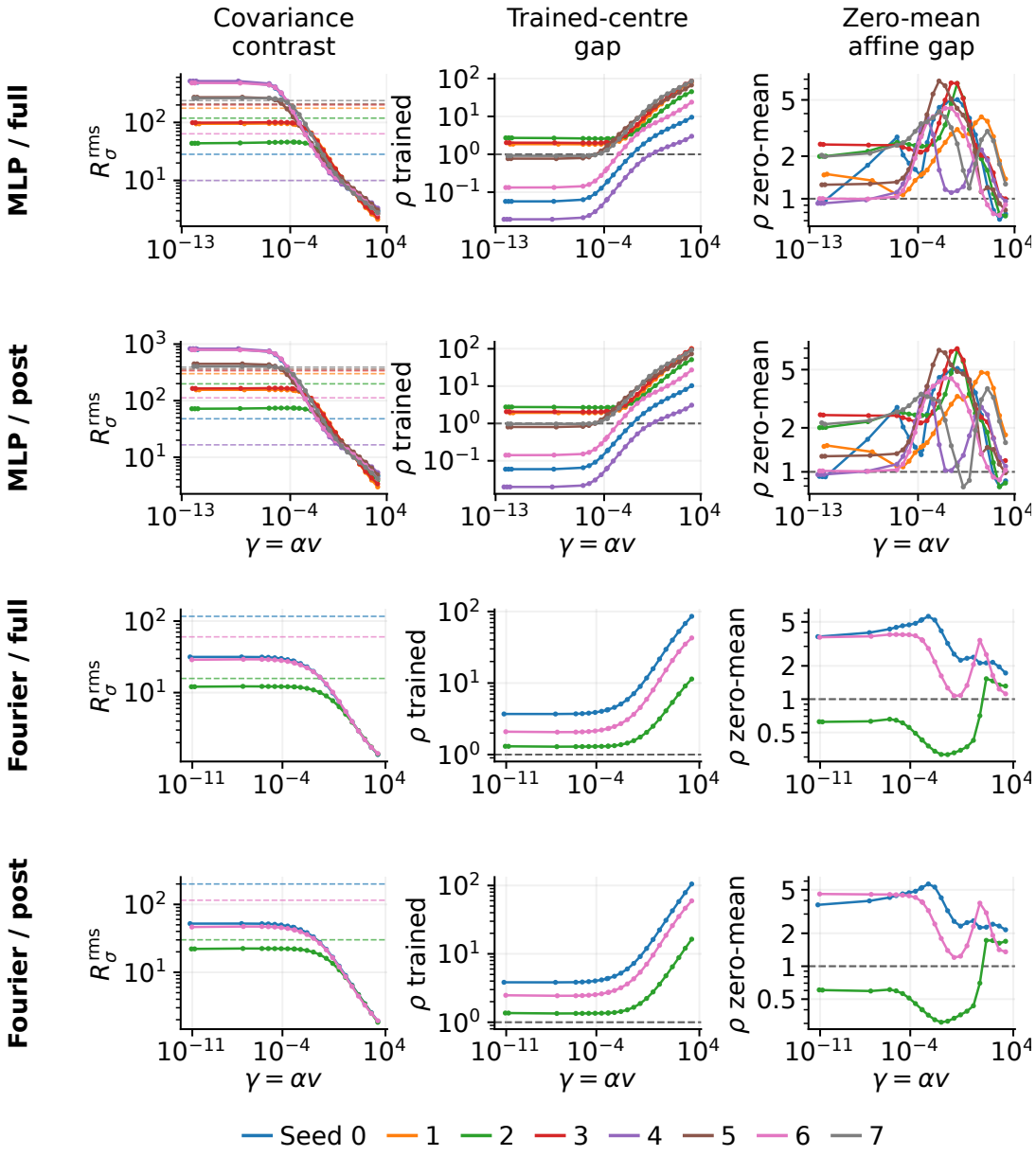


Figure 5: Burgers: eight admitted MLP and three admitted Fourier runs. Rows show MLP full-grid/post-formation, then Fourier full-grid/post-formation; post-formation intersects unchanged global masks with $t \geq 1/\pi$. Columns show RMS covariance contrast, trained gap and zero-mean affine gap; each gap uses its centre's error target. Coloured dashed lines give trained error targets; grey lines mark $\rho = 1$. Every distinct saved γ, including extensions and sensitivity aliases, uses the original Jacobian. Lines join samples without establishing monotonicity or a weak limit.

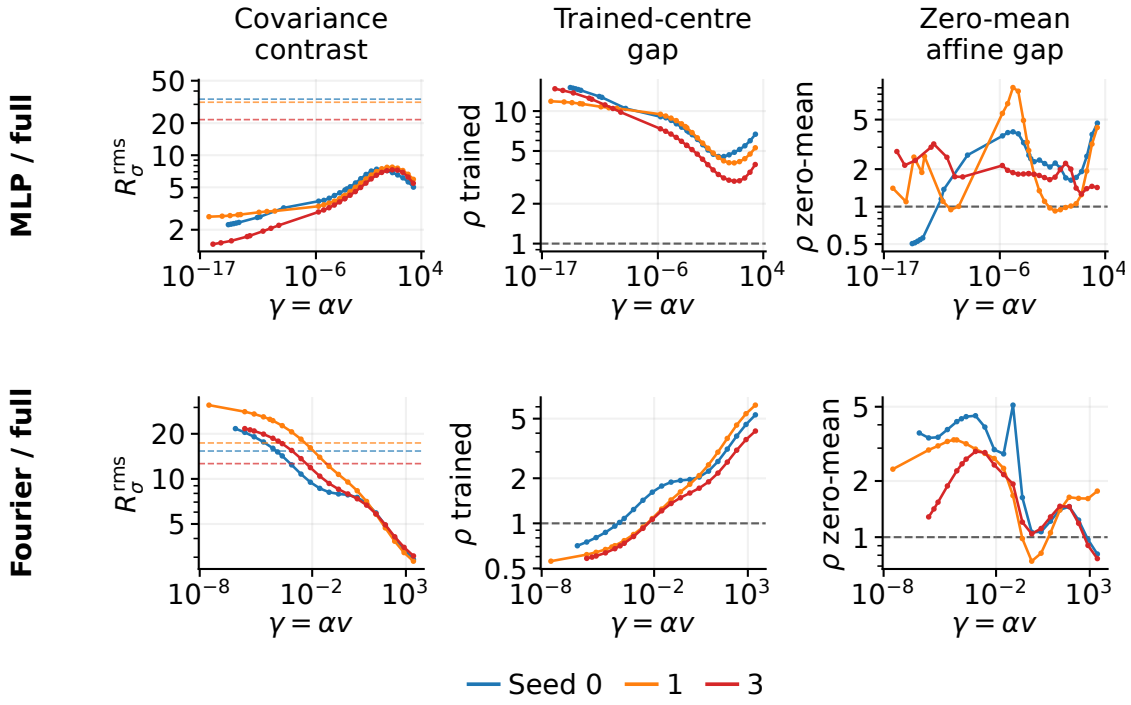


Figure 6: Allen–Cahn full-grid scale and centre curves for three admitted MLP (top) and three admitted Fourier (bottom) checkpoints. Columns and dashed lines follow Figure 5; each gap uses its centre's error target. Evidence settings for both families are in Table 14. All panels use the original Jacobian; lines connect saved settings.

family	region	baseline	MLE	original evidence	extended evidence
Burgers–MLP	bulk	[7.54, 54.1]	[0.264, 1.51]	[0.338, 1.53]	[0.349, 1.49]
	transition	[9.07, 109]	[0.288, 1.69]	[0.266, 1.53]	[0.286, 1.42]
	sharp	[0.58, 57.7]	[0.108, 8.55]	[0.114, 14.9]	[0.122, 16.9]
Burgers–Fourier	bulk	[3.87, 8.31]	[0.823, 2.42]	[0.714, 1.97]	[0.679, 1.84]
	transition	[3.51, 7.08]	[0.955, 1.8]	[0.807, 1.53]	[0.804, 1.42]
	sharp	[0.666, 0.864]	[0.167, 0.809]	[0.156, 0.695]	[0.155, 0.762]
Allen–Cahn–MLP	bulk	[15.7, 23]	[1.57, 2.59]	[2.43, 10.9]	[21.8, 30.5]
	transition	[10.8, 13.9]	[0.868, 1.64]	[1.18, 3.32]	[4.05, 21.2]
	sharp	[5.46, 7.22]	[0.416, 0.628]	[0.626, 0.81]	[0.626, 5.07]
Allen–Cahn–Fourier	bulk	[8.94, 9.95]	[0.97, 1.09]	[0.748, 0.963]	[0.748, 0.963]
	transition	[9.86, 11]	[0.794, 1.74]	[0.631, 1.13]	[0.631, 1.13]
	sharp	[4.38, 6.25]	[0.36, 2.26]	[0.331, 1.45]	[0.331, 1.45]

Table 17: Regional RMS zero-mean-prior affine displacement divided by trained error RMS on the same full-time region. Entries are across-run ranges. Baseline and MLE use $(\alpha_0, 1)$ and $(\alpha_0, \widehat{v})$; evidence columns use best tested grid pairs. Ratios above one indicate larger displacement than trained error, without determining accuracy.

Appendix E. Computation and Numerical Validation

This record specifies the checks supporting Appendix C; their reported defects and interpretation remain separate from forward-accuracy certification.

E.1 Direct-J Evaluation and Consistency Checks

The factorisation, grids and zero-mean-prior results are in Appendix C. At all 401 settings, automatic differentiation independently checks

$$h_\gamma = J^\top(r + J\delta_\gamma) + \gamma(\theta + \delta_\gamma),$$
$$\eta_{\text{back}} = \frac{\|h_\gamma\|}{(\lambda_{\max} + \gamma)\|\delta_\gamma\| + \|J^\top r + \gamma\theta\|}.$$

Absolute residuals, denominators and shift norms are archived. Complement checks use $J\delta_c$, norm, projection and output for $\delta_c = -Q_c Q_c^\top \theta$. Normwise residuals can miss weak-direction errors; null membership alone establishes neither amplitude nor completeness.

The 95 distinct θ-centred checkpoint–precision pairs merge repetitions among baseline, conditional MLE, original/extended evidence under both priors, and five trained-centre roots. Replacing h_γ by $J^\top(r + J\delta_L) + \gamma\delta_L$ and its denominator by $(\lambda_{\max} + \gamma)\|\delta_L\| + \|J^\top r\|$ gives maximum ratio 9.12×10^{-17}. Nonlinear evaluation retains J, g and covariance at θ; independent NumPy predictions at nine points per setting agree within 4.72×10^{-12} absolute.

Fields reproduce regional metrics and spectral sums reproduce mean variances. All 34 zero-mean-prior evidence selections and one-log-unit sets ($\ell \geq \max_{\text{grid}} \ell - 1$) reproduce from saved surfaces; all 34 θ-centred selections reproduce from corresponding factors. Saved arrays reproduce the 401 displacement norms and regional variances (Appendix C). Envelopes contain every tested intermediate variance field and a prescribed-spectrum null-mode example at 49 interior precisions; direct coverage at the five refined roots also lies within its envelopes. These consistency checks do not certify the original Jacobian by interval arithmetic.

E.2 Evidence Sensitivity Protocol

The six cases in Appendix C retain singular vectors and change the observation coherently by $z' = z + (s' - s)\theta_{\text{modes}}$, where $z = V^\top(J\theta - r)$ and $\theta_{\text{modes}} = U^\top Q_1^\top \theta$. An assumed singular-value budget gives

$$|\Delta\lambda_i| \leq 2s_i|\Delta s_i| + |\Delta s_i|^2.$$

At $s_i = \sqrt{\gamma}$, the relative envelope is $2\delta s/\sqrt{\gamma} + \delta s^2/\gamma$, with terms multiplied by 10 and 100 in the $10\times$ case. The exposed Allen–Cahn MLP quadratic terms are reported in Table 14. Formed-Gram roundoff (Table 5) is not a direct-J error bound.

All Allen–Cahn Fourier selections and Burgers–Fourier seeds $0, 6$ are unchanged; seed 2 moves noise and γ up one quarter decade in the increase-$10\times$ case. Every Burgers–MLP selection changes in at least two cases, retaining its noise boundary, with alpha ratios 0.562–3.162. Allen–Cahn MLP changes in all six cases: noise ratios reach 177.8 (seed 0) and 3162 (seeds $1, 3$), and γ ratios reach 1000 and $316{,}228$, respectively. Fixed-pair log-evidence changes reach about -1.05×10^6. The Allen–Cahn MLP scales are not robust estimates; fields at alternative selections use the original Jacobian.

The derivative in (10) groups modes by $\lambda_i/\gamma < 0.01$, 0.01–100 and > 100. For an isolated weak cluster, its leading log-evidence contribution is $-\frac{1}{2}[m \log v + \sum_i z_i^2/v]$ plus a constant, suggesting $v_* \approx m^{-1} \sum_i z_i^2$ at fixed α if all modes remain weak. For exact-null modes with positive projected energy this expression is exact. Positive modes can leave the cluster as v falls; other modes affect the full optimum. The grouped energies therefore explain a local boundary tendency, without a goodness-of-fit test, an alpha-reoptimised path or a $v \to 0$ conclusion. Allen–Cahn Fourier has positive transition-group contributions partly offsetting its negative weak group. Calculations retain the full formula; individual coordinates within unresolved clusters are not interpreted.

Against a local central log-noise finite difference, the largest slope defect is 4.43×10^{-6} absolute or 5.51×10^{-9} after scaling by $1 + |D_v|$. This verifies the calculation rather than evidence-model adequacy.

E.3 Synthetic Controls

The small 12×20 design compares QR/SVD posterior means, covariances and evidence with dense calculations, with maximum discrepancies 4.90×10^{-12} and 3.01×10^{-12} in relative norm and 1.42×10^{-13} in absolute log evidence. In 1000 correctly specified prior/noise draws, the affine posterior output mean gives one-sigma coverage 0.6870 (draw-level SE 0.00544) and 95% coverage 0.9534 (SE 0.00249). A separate full-column-rank least-squares example with $N = 40, P = 12$ and generating noise variance $v = 0.2$ gives a mean of 0.140411 (SE 0.001198) for the fitted-residual estimator $\|r_{\text{fit}}\|^2/N$, consistent with $(N - P)v/N = 0.14$. In a general linear fit the factor is $(N - \operatorname{rank}(X))/N$, reducing to $(N - P)/N$ only for full column rank. For a full-row-rank interpolating design it is zero; $P > N$ alone neither establishes that rank nor supplies a correction for nonlinear PINN fitting. Fitting can therefore reduce this estimator even under a correct noise model; this example is not an estimate of PINN discrepancy.

The fixed 100-draw overparameterised 16-decade eigenvalue toy has $100/100$ interior selections agreeing with its prescribed-spectrum reference. Hats in this toy denote evidence-selected grid values. At the $5, 25, 50, 75, 95$ percentiles, the selected log-scale ratios $\log_{10}(\widehat{\alpha}/\alpha)$ are $(0, 0, 0, 0, 0.25)$; $\log_{10}(\widehat{v}/v)$ and $\log_{10}(\widehat{\gamma}/\gamma)$ are $(-0.25, 0, 0, 0, 0.25)$. These quarter-decade-grid medians coincide with the generating pair without establishing exact scale recovery.

A 21.2-decade synthetic spectrum with $N = 96, P = 160$ has $79/96$ prescribed eigenvalues below $\varepsilon_{64}\lambda_{\max}$. Its generating $\gamma = 10^{-8}$ is interior to the search grid. All 100 selected pairs are interior and agree with the prescribed-spectrum reference. The same log-scale percentiles are $(-0.25, 0, 0, 0, 0.25)$ for α and γ, and $(0, 0, 0, 0, 0)$ for v. The maximum selected log-evidence discrepancy is 9.40×10^{-8}. Across the whole tested grid, however, the maximum absolute discrepancy is 0.01775 and the maximum scaled discrepancy $|\ell_{\text{num}} - \ell_{\text{ref}}|/(1 + |\ell_{\text{ref}}|)$ is 4.55×10^{-5}, exceeding the prescribed 10^{-7} tolerance. Here ℓ_{num} and ℓ_{ref} include the Gaussian normalising constant. Near-best discrepancies are small. Cancellation and the additive constant affect this scaled error; the failed whole-grid check alone establishes neither an incorrect selector nor model inadequacy. These controls retain the original tolerance and do not establish PINN evidence stability.

Appendix F. Formal Development

The independent `PaperReliability` development uses Lean `v4.30.0` (de Moura and Ullrich, 2021) and the matching Mathlib release (The mathlib Community, 2020). Table 18 records its analytic scope. The covariance bridge assumes a supplied exact orthogonal spectral representation.

Physical modal applications use nonnegative eigenvalues and energies, positive precision and the stated positive pointwise or bulk denominators. Kernel statements concern exact zero modes, including the whole kernel rather than a numerical rank cutoff. Coverage envelopes use fixed trained errors and regions, an actual empirical higher quantile, and a precision supplied within the retained bracket.

manuscript result	Lean declarations
Matrix-to-modal covariance, (3)	`spectral_covariance_representation`
Fixed-cutoff contrast bounds, (7)	`two_subspace_contrast_enclosure`
Regional variance aggregation, (6)	`regional_two_subspace_enclosure`
Sharpness witnesses, Proposition 9	`upper_envelope_arbitrary_energy_sharpness` `lower_envelope_arbitrary_energy_sharpness`
Conditional modal limits, Section 3.3	`regional_contrast_strong_limit` `regional_contrast_weak_limit`
Nonmonotone counterexample, Section 3.3	`endpoint_limits_do_not_bound_contrast`
Exact-kernel prior attribution, Appendix C	`kernel_prediction_shift_attribution`
Pointwise likelihood-shift bound, (11)	`likelihood_predictive_shift_bound`
Regional prior-centre bounds, (12)	`likelihood_regional_error_bounds`
Prior-centre coverage brackets, Appendix C	`likelihood_centre_coverage`
Positive-modal and retained-bracket envelopes, (13)	`positive_modal_variance_enclosure` `variance_enclosure_on_precision_bracket`
Bulk-calibrated sharp coverage, (14)	`calibrated_modal_coverage_enclosure`

Table 18: The `PaperReliability` library exposes 15 public analytic declarations, all free of `sorry`. Private proof lemmas provide supporting infrastructure.

Verification covers fixed-cutoff inequalities and sharpness witnesses with prescribed energies in spectra bounded by $\lambda_{\max}$ and explicit approach factors as $\epsilon' \downarrow \epsilon$. Maximum-eigenvalue equalities and supremum/infimum statements are not separate formal results. The expansion (4), secondary mean-width identities and transfers, Remarks 5–6, and spectral-decomposition existence remain unverified. The proofs concern real-model inequalities, not QR/SVD accuracy, saved values, run counts or numerical root brackets.

References

J. Antorán, D. Janz, J. U. Allingham, E. Daxberger, R. Barbano, E. Nalisnick, and J. M. Hernández-Lobato. Adapting the linearised Laplace model evidence for modern deep learning. In *International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 796–821, 2022.

A. Daw, J. Bu, S. Wang, P. Perdikaris, and A. Karpatne. Mitigating propagation failures in physics-informed neural networks using retain-resample-release (R3) sampling. In *International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 7264–7302, 2023.

E. Daxberger, A. Kristiadi, A. Immer, R. Eschenhagen, M. Bauer, and P. Hennig. Laplace Redux – effortless Bayesian deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 20089–20103, 2021.

L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In A. Platzer and G. Sutcliffe, editors, *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

P. Flores, O. Graf, P. Protopapas, and K. Pichara. Improved uncertainty quantification in physics-informed neural networks using error bounds and solution bundles. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 286 of *Proceedings of Machine Learning Research*, pages 1289–1336, 2025.

V. Gopakumar, A. Gray, L. Zanisi, T. Nunn, D. Giles, M. Kusner, S. Pamela, and M. P. Deisenroth. Calibrated physics-informed uncertainty quantification. In *International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 20103–20141, 2025.

K. M. Graczyk and K. Witkowski. Bayesian reasoning for physics-informed neural networks. *Physical Review E*, 113:055307, 2026. doi: 10.1103/29bd-jfhz.

O. Graf, P. Flores, P. Protopapas, and K. Pichara. Error-aware B-PINNs: Improving uncertainty quantification in Bayesian physics-informed neural networks. *arXiv preprint arXiv:2212.06965*, 2022.

A. Immer, M. Bauer, V. Fortuin, G. Rätsch, and M. E. Khan. Scalable marginal likelihood estimation for model selection in deep learning. In *International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 4563–4573, 2021a.

A. Immer, M. Korzepa, and M. Bauer. Improving predictions of Bayesian neural nets via local linearization. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 130 of *Proceedings of Machine Learning Research*, pages 703–711, 2021b.

M. Izzatullah, I. E. Yildirim, U. B. Waheed, and T. Alkhalifah. Laplace HypoPINN: Physics-informed neural network for hypocenter localization and its predictive uncertainty. *Machine Learning: Science and Technology*, 3(4):045001, 2022. doi: 10.1088/2632-2153/ac94b3.

A.-K. Kassam and L. N. Trefethen. Fourth-order time-stepping for stiff PDEs. *SIAM Journal on Scientific Computing*, 26(4):1214–1233, 2005.

D. P. Kingma and J. L. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. URL `https://arxiv.org/abs/1412.6980`.

A. S. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby, and M. W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 26548–26560, 2021.

A. Kristiadi, M. Hein, and P. Hennig. Learnable uncertainty under Laplace approximations. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 161 of *Proceedings of Machine Learning Research*, pages 344–353, 2021.

B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 6402–6413, 2017. URL `https://proceedings.neurips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html`.

J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018. doi: 10.1080/01621459.2017.1307116.

K. Levenberg. A method for the solution of certain non-linear problems in least squares. *Quarterly of Applied Mathematics*, 2(2):164–168, 1944. doi: 10.1090/qam/10666.

D. J. C. MacKay. Bayesian interpolation. *Neural Computation*, 4(3):415–447, 1992. doi: 10.1162/neco.1992.4.3.415.

E. Magnani, M. Pförtner, T. Weber, and P. Hennig. Linearization turns neural operators into function-valued Gaussian processes. In *International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 42421–42452, 2025.

D. W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963. doi: 10.1137/0111030.

S. Mishra and R. Molinaro. Estimates on the generalization error of physics-informed neural networks for approximating PDEs. *IMA Journal of Numerical Analysis*, 43(1):1–43, 2023. doi: 10.1093/imanum/drab093.

A. F. Psaros, X. Meng, Z. Zou, L. Guo, and G. E. Karniadakis. Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons. *Journal of Computational Physics*, 477:111902, 2023.

M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

P. Rathore, W. Lei, Z. Frangella, L. Lu, and M. Udell. Challenges in training PINNs: A loss landscape perspective. In *International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 42159–42191, 2024.

M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 7537–7547, 2020.

The mathlib Community. The Lean mathematical library. In *ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824.

S. Wang, Y. Teng, and P. Perdikaris. Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(5):A3055–A3081, 2021.

S. Wang, X. Yu, and P. Perdikaris. When and why PINNs fail to train: A neural tangent kernel perspective. *Journal of Computational Physics*, 449:110768, 2022.

L. Yang, X. Meng, and G. E. Karniadakis. B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021.

Y. Yu, C. H. Ho, and Y. Wang. A conformal prediction framework for uncertainty quantification in physics-informed neural networks. *Journal of Computational Physics*, 561:114979, 2026. doi: 10.1016/j.jcp.2026.114979.